Straight Line

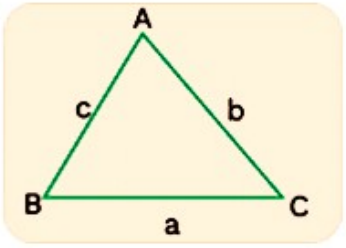
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Straight Line

Triangle :

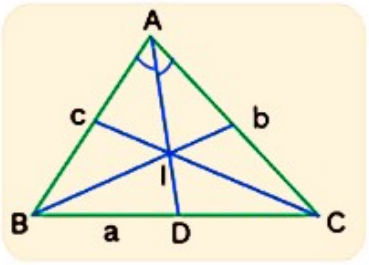
$\angle B > \angle C \Rightarrow b > c$
$\angle A + \angle B + \angle C = \pi$
$a + b > c$
$|a - b| < c$



Angle Bisector :

$$AD = \frac{2bc}{b+c}\cos\left(\frac{A}{2}\right)$$

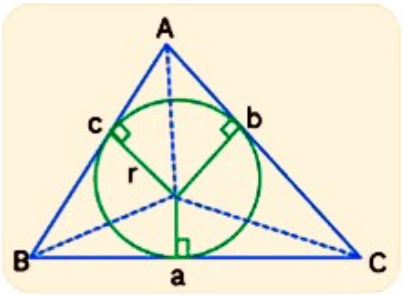
I is called Incentre
(Point of concurrency of internal angle bisector)



Incircle :

Circle which touches sides of triangle is called incircle.

$$r = \frac{\Delta}{s}$$

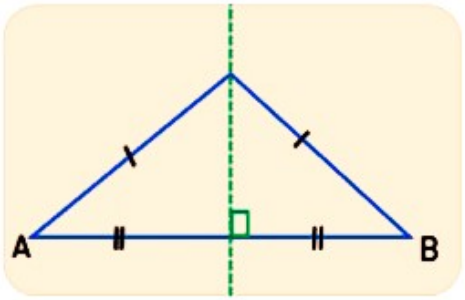


Altitude :

Perpendicular from vertex to opposite side (orthocentre)

Medium :

Line joining vertex to mid-point of opposite sides (centroid)



Perpendicular Bisector :

Any point on perpendicular bisector is at equal distance from A & B.

Quadrilaterals:

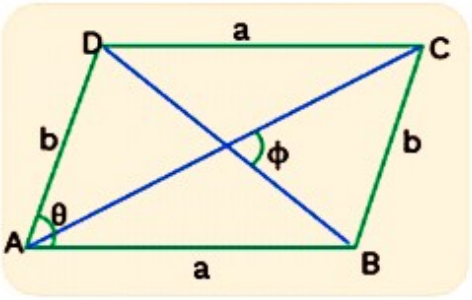
Sum of all interior angles of n-sided figure is = $(n - 2)\pi$

Parallelogram :

(i) Opposite sides are parallel and equal

(ii) Adjacent angle are supplementary

(iii) Diagonals are bisected



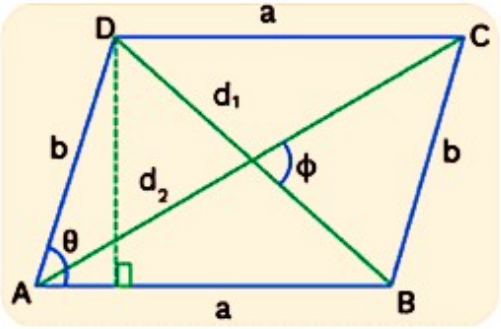
Parallelogram :

Area of parallelogram

$$= \frac{1}{2} d_1 d_2 \sin\phi$$

DE = $b\sin\theta$

Area of parallelogram = $a\, b\sin\theta$



Rhombus :

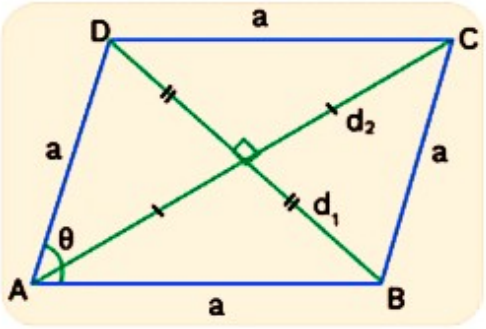
Parallelogram will be Rhombus if

(i) Diagonals are perpendicular

(ii) sides equal

(iii) Diagonals bisect the angles of parallelogram

(iv) Area of Rhombus = $\frac{1}{2} d_1 d_2$



Rectangle :

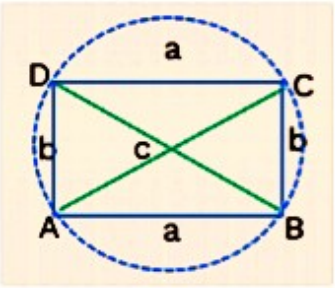
Parallelogram will be rectangle if

(i) Angle 90°

(ii) Diagonals are equal

(iii) $a^2 + b^2 = c^2$

(iv) Rectangle is cyclic quadrilateral



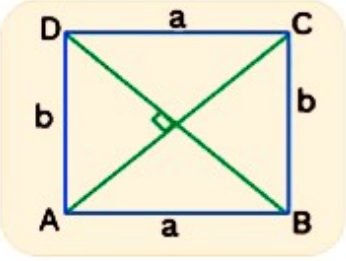
Square :

A Rectangle will be square if

(i) Sides equal

(ii) Diagonals are perpendicular

(iii) Diagonals are angle bisector

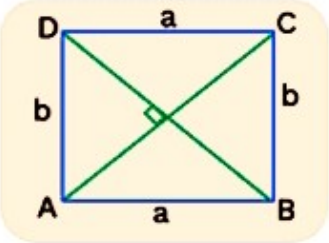


Note :

A Rhombus will be square if

(i) Diagonals are equal

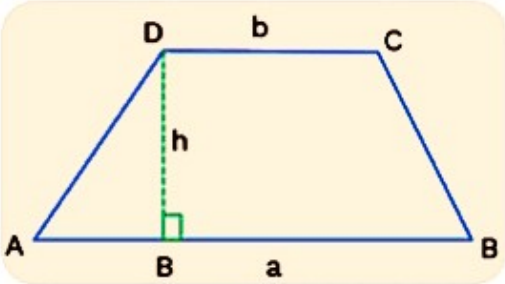
(ii) Angles 90°



Trapezium :

One pair of opposite sides are parallel

$$\text{Area} = \frac{1}{2}(a+b)\times h$$

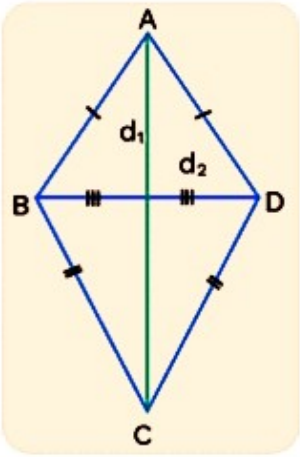


Kite :

(i) One diagonal divide figure into two congruent parts.

(ii) Diagonals are perpendicular

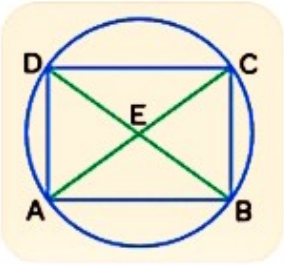
(iii) $\text{Area} = \frac{1}{2}d_1 \times d_2$



Cyclic Quadrilateral :

(i) Vertices lie on circle

(ii) A + C = π = B + D

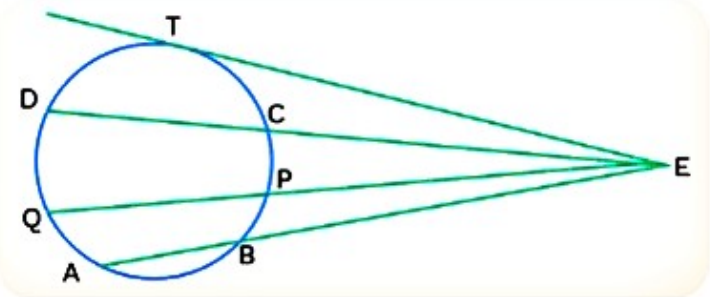


Note :

EB. EA = EC. ED

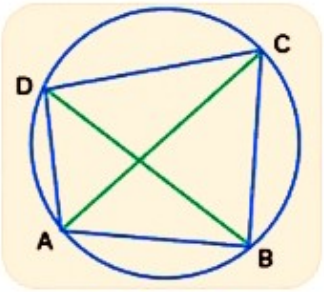
$= ET^2$

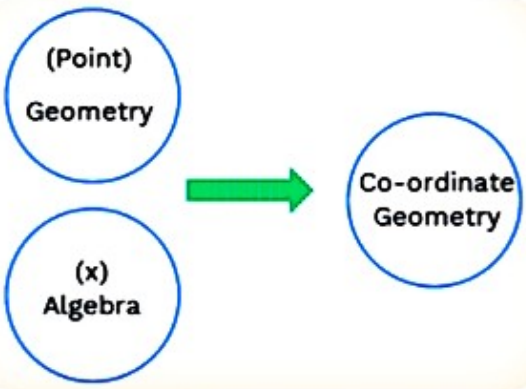
= EP. EQ



Ptolmey's Theorem :

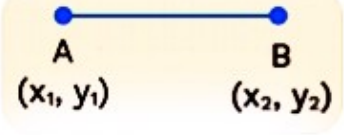
Sum of product of opposite side = Product of diagonals





Distance Formulae

$$AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



Q. **Find distance between following points:**

(i) (1,3), (4, –1) **(ii) (0,0), (–5, –12)** **(iii) (1,1), (16, 9)**

(iv) (0,0), (40,9) **(v) (0,0), (2cosθ, 2sinθ)**

Sol. (i) Distance = $\sqrt{(1-4)^2 + (3+1)^2} = \sqrt{9+16} = \sqrt{25}$ = 5 unit

(ii) Distance = $\sqrt{(0+5)^2 + (0+12)^2} = \sqrt{25+144} = \sqrt{169}$ = 13 unit

(iii) Distance = $\sqrt{(1-16)^2 + (1-9)^2} = \sqrt{225+64} = \sqrt{289}$ = 17 units

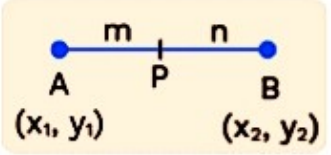
(iv) Distance = $\sqrt{(0-40)^2 + (0-9)^2} = \sqrt{1600+81} = \sqrt{1681}$ = 41 unit

(v) Distance = $\sqrt{(0-2\cos\theta)^2 + (0-2\sin\theta)^2} = \sqrt{4\cos^2\theta + 4\sin^2\theta} = \sqrt{4}$ = 2 unit

Section Formulae (Internal Division)

If two point $A(x_1, y_1)$ and $B(x_2, y_2)$ are divided by point P in ratio m : n, then coordinate of P will be

$$P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

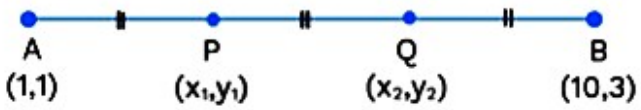


Coordinate of mid point of $A(x_1, y_1)$, $B(x_2, y_2)$

$$P = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

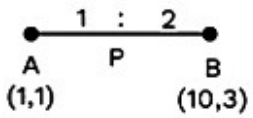
Q. **Find points of trisection of (1,1) and (10,3).**

Sol. Let P, Q points trisects the line segment AB Now for coordinate of P

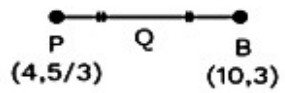


$$P\left(\frac{1\times 10+2\times 1}{1+2},\frac{1\times 3+2\times 1}{1+2}\right)$$

$$P\left(4,\frac{5}{3}\right)\equiv(x_1,y_1)$$



Now for coordinate of Q
∵ Q is mid point of PB
hence,

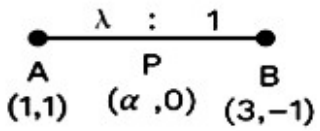


$$Q\left(\frac{4+10}{2},\frac{\frac{5}{3}+3}{2}\right)$$

$$Q\left(7,\frac{7}{3}\right)\equiv(x_2,y_2)$$

Q. **Find the ratio in which point on x-axis divides the two points (1,1), (3,–1) internally.**

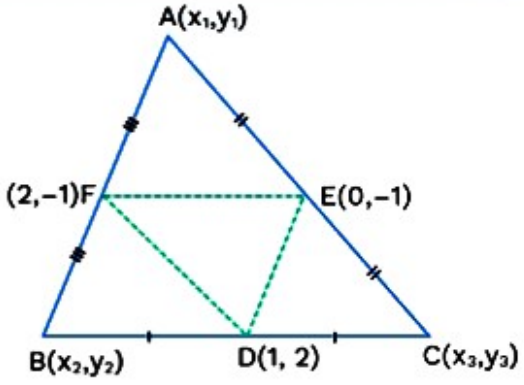
Sol. Let point $P(\alpha,0)$ on x-axis divides the line segment joining A(1, 1) and B(3, –1) in ratio $\lambda:1$



Hence, $\alpha=\frac{\lambda\times 3+1\times 1}{\lambda+1}, 0=\frac{\lambda(-1)+1\times 1}{\lambda+1}\Rightarrow\lambda=1$

Q. **Mid point of sides of a triangle are (1,2),(0,–1) and (2,–1). Find the co-ordinate of vertices.**

Sol. Given D(1, 2), E(0, –1) and F(2, –1)
Now,
AEDF is a parallelogram.
Hence, mid-point of AD and EF is same
hence,



$$\frac{x_1+1}{2}=\frac{2+0}{2}\Rightarrow x_1=2+0-1=1$$

$\frac{y_1+2}{2} = \frac{-1-1}{2} \Rightarrow y_1 = -1-1-2 = -4$

Similarly $x_2 = 2+1-0$, $y_2 = -1+2+1 \Rightarrow B(3,2)$

$x_3 = 1+0-2$, $y_3 = 2-1+1 \Rightarrow C(-1,2)$

Q. **Coordinate A, B, C are (4, 1), (5, –2) and (3, 7). Find D so that A,B,C,D is a parallelogram.**

Sol.

Case-I:

$\alpha = 4+3-5 = 2$

$\beta = 1+7+2 = 10$

Hence, D(2, 10)

Case-II:

$x = 4+5-3$

$= 6$

$y = 1-2-7$

$= -8$

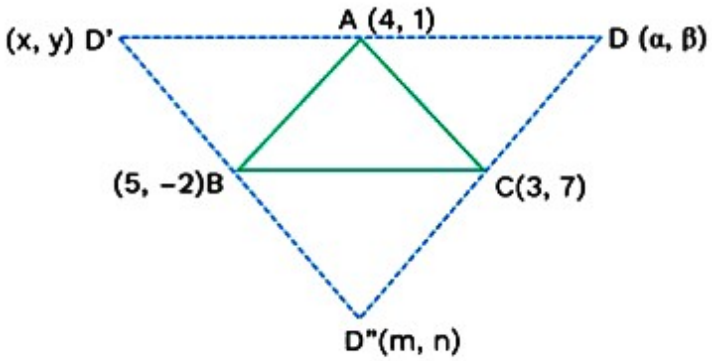
Hence, D'(6,–8)

Case-III:

$m = 5+3-4 = 4$

$n = -2+7-1 = 4$

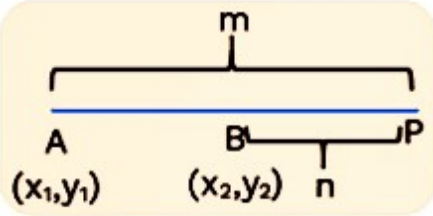
Hence, D"(4, 4)



Section Formulae (External Division)

If two point $A(x_1, y_1)$ and $B(x_2, y_2)$ are divided by point P in ratio m : n (externally), then coordinate of P will be

$$P\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$$

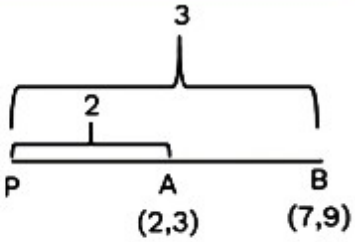


Q. **Find the point dividing (2,3), (7,9) externally in the ratio 2 : 3**

Sol.

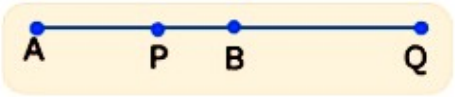
$$P\left(\frac{2\times 7 - 3\times 2}{2-3}, \frac{2\times 9 - 3\times 3}{2-3}\right)$$

P(–8,–9)



Harmonic Conjugate

If a point P divides AB internally in the ratio a : b and point Q divides AB externally in the ratio a : b then P and Q are said to be harmonic conjugate of each other w.r.t. AB



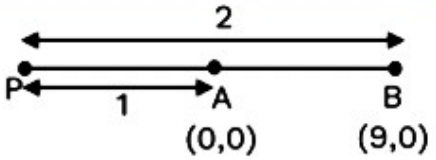
$$\frac{AP}{BP} = \frac{\lambda}{1} = \frac{AQ}{BQ}$$

Harmonic Mean

$$AB = \frac{2AP.AQ}{AP + AQ}$$

Q. **If coordinate of A & B is (0, 0) and (9,0). Find point which divide AB externally in the ratio 1 : 2. Find its harmonic conjugate.**

Sol. Let point P divides AB in 1 : 2 externally

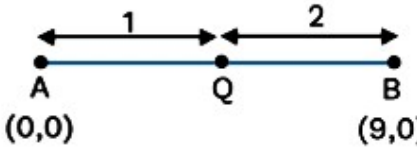


then $P\left(\frac{1(9)-2(0)}{1-2}, \frac{1(0)-2(0)}{1-2}\right)$

P(−9, 0)

Now Let Q is harmonic conjugate of P w.r.t AB

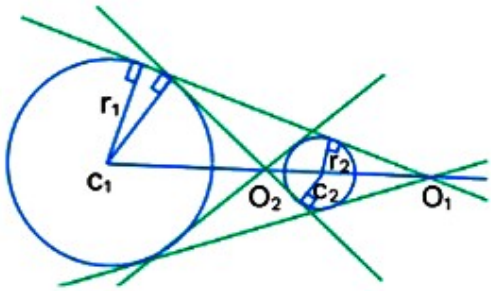
Hence, Q divides AB in 1 : 2 internally



$$Q\left(\frac{1(9)+2(0)}{1+2}, \frac{1(0)+2(0)}{1+2}\right)$$

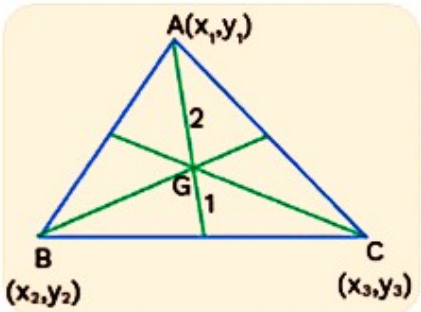
Q(3, 0)

External and Internal common tangents divide line segment joining centre of two circles externally and internally at O_1 and O_2 is $\frac{r_1}{r_2}$.



Co-ordinate of G

Centroid is point of intersection of medians and its coordinates will be given by



$$G \equiv \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

Properties of centroid :

(i) All medians divide given triangle in 6 equal parts.

(ii) If AD is the median through A with G on it then $\frac{AG}{GD}=\frac{2}{1}$, where D lies on BC.

Similarly, $\frac{BG}{GE}=\frac{2}{1}$, where E is the mid-point of AC.

(iii) Centroid of triangles obtained by joining mid-points will be same as of original triangle.

Q. **Find mid points of sides of Δ if vertices are given (0,0), (2,3), (4,0) also find coordinate of G**

Sol. Mid point of BC

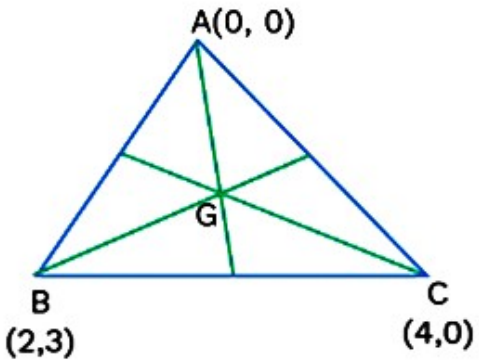
$$D\left(\frac{2+4}{2},\frac{3+0}{2}\right)\equiv\left(3,\frac{3}{2}\right)$$

Mid point of AC

$$E\left(\frac{0+4}{2},\frac{0+0}{2}\right)\equiv(2,0)$$

Mid point of AB, $F\left(\frac{0+2}{2},\frac{0+3}{2}\right)\equiv\left(1,\frac{3}{2}\right)$

also $G\left(\frac{0+2+4}{3},\frac{0+3+0}{3}\right)\equiv G(2,1)$



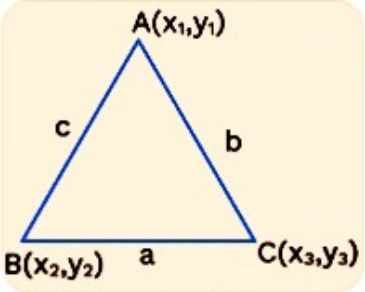
Coordinate of Incentre (I)

Incentre is the point of intersection of angle bisectors.

For the triangle ABC, where $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ the coordinate of incentre will be

$$I\left(\frac{ax_1+bx_2+cx_3}{a+b+c},\frac{ay_1+by_2+cy_3}{a+b+c}\right)$$

where a, b, c are the length of sides of triangle opposite to vertex A, B and C.



Properties of Incentre

(i) If angle bisector from A intersects BC at point D then $\frac{BD}{DC}=\frac{c}{b}$.

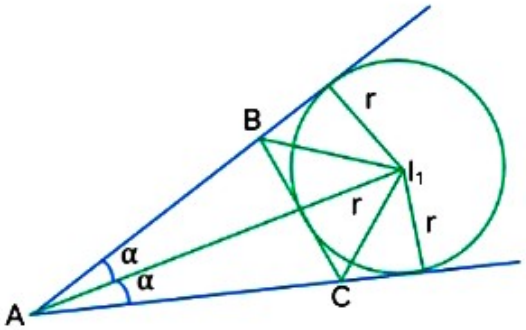
(ii) If AD is the bisector of angle A with I (incentre) lying on it then $\frac{AI}{ID}=\frac{b+c}{a}$

(iii) Image of A along internal bisector of C and B will lie on BC.

Ex-centre :

The point of intersection of the external bisectors of the angles of a triangle is called its ex-centre.

The circle opposite to the vertex A is called the escribed circle opposite A or the circle excribed to the side B C. If I_1 is the point of intersection of internal bisector of ∠BAC and external bisector of ∠ABC and ∠ACB, then



$$I_1=\left(\frac{-ax_1+bx_2+cx_3}{-a+b+c},\frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$$

$$I_2=\left(\frac{ax_1-bx_2+cx_3}{a-b+c},\frac{ay_1-by_2+cy_3}{a-b+c}\right)$$

and $I_3=\left(\frac{ax_1+bx_2-cx_3}{a+b-c},\frac{ay_1+by_2-cy_3}{a+b-c}\right)$

Respectively.

Q. **Line 3x + 4y = 12, x = 0, y = 0 form a Δ. Find the centre of circles touching the line and the coordinate axis.**

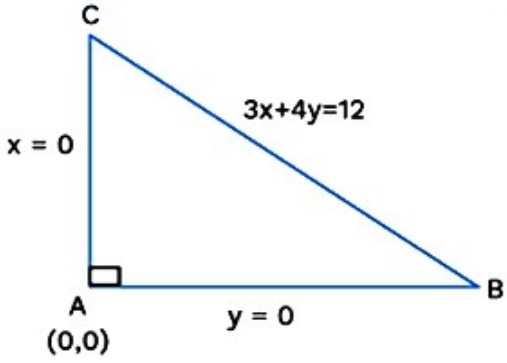
Sol. Coordinate of B(4, 0)

and C(0, 3)

Now a = BC = 5

b = AC = 3

c = AB = 4



Incircle

$$\text{Centre } I \equiv \left(\frac{5(0)+3(4)+4(0)}{5+3+4}, \frac{5(0)+3(0)+4(3)}{5+3+4}\right)$$

$I = (1, 1)$

Excircle

$$I_1 \equiv \left(\frac{-5(0)+3(4)+4(0)}{-5+3+4}, \frac{-5(0)+3(0)+4(3)}{-5+3+4}\right)$$

$I_1 = (6, 6)$

$$I_2 \equiv \left(\frac{5(0)-3(4)+4(0)}{5-3+4}, \frac{5(0)-3(0)+4(3)}{5-3+4}\right)$$

$I_2 = (-2, +2)$

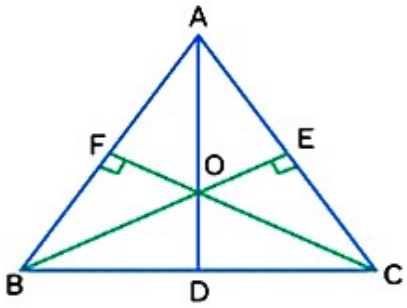
$$I_3 \equiv \left(\frac{5(0)+3(4)-4(0)}{5+3-4}, \frac{5(0)+3(0)-4(3)}{5+3-4}\right)$$

$I_3 = (3, -3)$

Circumcentre:

The point of intersection of the perpendicular bisectors of a triangle is called its circumcentre.

If the vertices of ΔABC be $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then the co-ordinates of its circumcentre is



$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

Notes:

(i) If O(x, y) be the circumcentre of ΔABC, its co-ordinates is determined by the relation $OA = OB = OC$.

(ii) In case of a right-angled triangle, mid-point of the hypotenuse is the circumcentre.

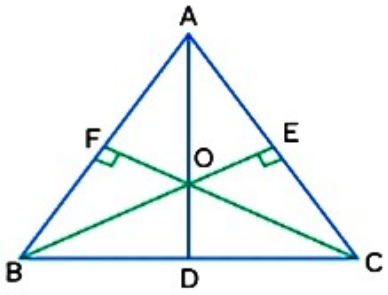
(iii) Circumcentre is equidistant from all the vertices of triangle.

Orthocentre:

The point of intersection of the altitudes of a triangle is called its orthocentre.

If the vertices of ΔABC be $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, the co-ordinates of its orthocentre is

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C}\right)$$



Properties of Orthocentre:

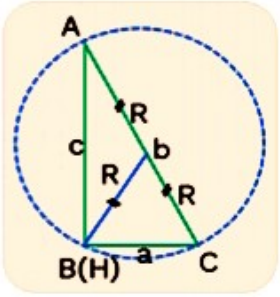
(i) If for triangle ABC, O is the orthocentre then for triangle ABO, C will be the orthocentre. Similarly, for triangle BOC, A will be the orthocentre.

(ii) Reflection of orthocentre with respective to any line of triangle will lie on circumcircle.

Notes:

(i) In an isosceles triangle, G, O, I and C lie on the same line and in an equilateral triangle, all these four points coincide.

(ii) The orthocentre of ΔABC are (a, b),(b, a) and (a, a) is (a, a).

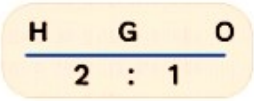


(iii) In case of an obtuse-angle triangle, the circumcentre and orhtocentre both lie outside of the triangle.

(iv) In Right angle triangle

H is orthocentre

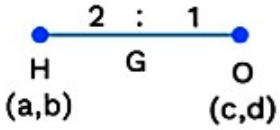
(v) G (centroid) and I (Incentre) always lies in interior of triangle whereas H (orthocentre) and O (circumcentre) lies on inside, outside or periphery depending upon triangle being acute, obtuse or right angle.



Q. **Orthocentre and circumcentre of a ΔABC are (a, b), (c, d). If the co-ordinate of the vertex A are (x_1, y_1). Then find coordinate of middle point of BC.**

Sol. Centroid

$$G\left(\frac{2c+a}{3}, \frac{2d+b}{3}\right)$$



Now, let mid point of BC is $D(\alpha, \beta)$. then

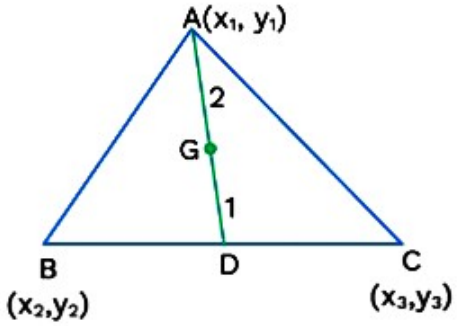
$\because$ G divides AD in 2:1 internally
hence,

$$G\left(\frac{2\alpha + x_1}{3}, \frac{2\beta + y_1}{3}\right)$$

$$\Rightarrow \frac{2c+a}{3} = \frac{2\alpha + x_1}{3}, \frac{2d+b}{3} = \frac{2\beta + y_1}{3}$$

$$\Rightarrow \alpha = \frac{1}{2}(2c + a - x_1), \ \beta = \frac{1}{2}(2d + b - y_1)$$

$$\Rightarrow D\left(\frac{1}{2}(2c + a - x_1), \frac{1}{2}(2d + b - y_1)\right)$$



Q. **Vertices of a triangle are (2, -2), (-2, 1), (5, 2). Find distance between circumcentre and centroid.**

Sol. Given A(2,–2), B(–2,1), C(5,2)

centroid $G\left(\frac{2-2+5}{3}, \frac{-2+1+2}{3}\right) = G\left(\frac{5}{3}, \frac{1}{3}\right)$

also $AB = \sqrt{4^2 + 3^2} = 5$

$BC = \sqrt{7^2 + 1^2} = \sqrt{50}$

$AC = \sqrt{3^2 + 4^2} = 5$

Clearly $AB^2 + AC^2 = BC^2$
$\Rightarrow$ ΔABC is right angled at A now circuncentre O will be mid point of BC hence

Circumcentre $O\left(\frac{-2+5}{2}, \frac{1+2}{2}\right)$

$$\Rightarrow O\left(\frac{3}{2}, \frac{3}{2}\right)$$

distance = OG = $\sqrt{\left(\frac{5}{3}-\frac{3}{2}\right)^2+\left(\frac{1}{3}-\frac{3}{2}\right)^2}$

$= \sqrt{\frac{1}{36}+\frac{49}{36}} = \frac{5}{3\sqrt{2}}$ unit

Q. **A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle.**

Sol. $\because 18^2 + 24^2 = 30^2$

$\Rightarrow$ triangle is right angled

let vertices are A(0,0), B(0,18) & C(24,0)

Now centroid $G\left(\frac{24}{3},\frac{18}{3}\right) \equiv G(8,6)$

Circumcentre O will be mid point of BC

Hence $O\left(\frac{24}{2},\frac{18}{2}\right) \equiv O(12,9)$

Incenter, $I\left(\frac{30(0)+24(0)+18(24)}{30+24+18},\frac{30(0)+24(18)+18(0)}{30+24+18}\right)$

I(6, 6)

for Δ IOG

sides $IO = \sqrt{6^2+3^2} = 3\sqrt{5}$

$OG = \sqrt{4^2+3^2} = 5$

$IG = \sqrt{2^2+0^2} = 2$

$s = \frac{7+3\sqrt{5}}{2}$

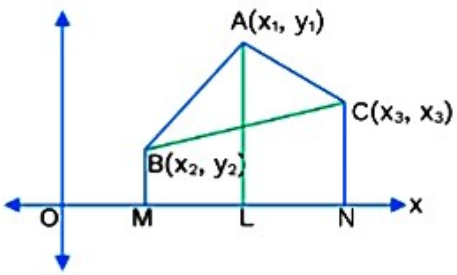
Area = $\sqrt{\left(\frac{7+3\sqrt{5}}{2}\right)\left(\frac{7-3\sqrt{5}}{2}\right)\left(\frac{3\sqrt{5}-3}{2}\right)\left(\frac{3\sqrt{5}+3}{2}\right)} = \sqrt{\frac{4}{4}\times\frac{36}{4}}$ = 3 unit.

Area of A triangle

If $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of ΔABC, its area is given by

$$\frac{1}{2}\left[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right]$$

or $$\frac{1}{2}\begin{Vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{Vmatrix}$$



or $\frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}$

Notes:

(i) The area of a polygon, with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3), ...,(x_n, y_n) is

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \cdots + \frac{1}{2}\begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix}.$$

(ii) If the vertices of a triangle be $A(r_1, \theta_1)$, $B(r_2, \theta_2)$ and (r_3, θ_3), its area is

$$\frac{1}{2}\left[r_1r_2\sin(\theta_1 - \theta_2) + r_2r_3\sin(\theta_2 - \theta_3) + r_3r_1\sin(\theta_3 - \theta_1)\right]$$

(iii) If $a_ix + b_iy + c_i = 0$, $i = 1, 2, 3$ be the sies of a triangle, its area is

$$\frac{1}{2C_1C_2C_3}\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2$$

Awhere C_1, C_2 and C_3 are the cofactors of c_1, c_2 and c_3 in the determinant $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$.

(iv) If $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a triangle, it area is given by

$$\frac{1}{2}\begin{vmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{vmatrix}.$$

(v) If $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be three collinear, points then

$$\begin{vmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{vmatrix} = 0$$

or

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

Condition of collinearity of $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$

$$\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = 0$$

Note:

All coordinates of the vertices of an equilateral Δ can't be integer.

Q. **Find k for which points (k + 1, 2 – k), (1 – k, –k), (2 + k, 3 – k) are collinear.**

Sol.

$$\begin{vmatrix} 1 & 1 & 1 \\ k+1 & 1-k & 2+k \\ 2-k & -k & 3-k \end{vmatrix} = 0$$

$C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$\begin{vmatrix} 0 & 0 & 1 \\ 2k & -1-2k & 2+k \\ 2 & -3 & 3-k \end{vmatrix} = 0$$

$\Rightarrow 1\{-6k - 2(-1-2k)\} = 0$

$-2k + 2 = 0 \Rightarrow k = 1$

Q. **If points (a, 0), (0, b) and (1, 1) are collinear, then prove that $\frac{1}{a} + \frac{1}{b} = 1$**

Sol. Clearly

$$\begin{vmatrix} 1 & 1 & 1 \\ a & 0 & 1 \\ 0 & b & 1 \end{vmatrix} = 0$$

$1(-b) - 1(a) + 1(ab) = 0$

$a + b = ab$

divide by ab $\Rightarrow \frac{1}{b} + \frac{1}{a} = 1$

Q. **Find relation between x & y if (x, y) lies on line joining the points (2, –3) and (1,4)**

Sol. Let a general point on line P(x, y)

Now all points are collinear.

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ x & 2 & 1 \\ y & -3 & 4 \end{vmatrix} = 0$$

$\Rightarrow 1(11) - 1(4x - y) + 1(-3x - 2y) = 0$

$\Rightarrow -7x - y + 11 = 0$

$\Rightarrow 7x + y = 11$ (required relation)

Q. **Show that (b, c + a), (c, a + b) and (a, b + c) are collinear.**

Sol.

$$\begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c+a & a+b & b+c \end{vmatrix}$$

$C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$\begin{vmatrix} 0 & 0 & 1 \\ b-c & c-a & a \\ c-b & a-c & b+c \end{vmatrix}$$

$$(b-c)(c-a)\begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & a \\ -1 & -1 & b+c \end{vmatrix} = 0$$

Q. **If the area of Δ formed by points (1, 2), (2, 3) and (x, 4) is 40 sq. units. Find x.**

Sol. Area = 40

$$\frac{1}{2}\left\|\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 2 & x \\ 3 & 4 \end{vmatrix} + \begin{vmatrix} x & 1 \\ 4 & 2 \end{vmatrix}\right\| = 0$$

$|-1 + (8-3x) + (2x-4)| = 80$

$|3-x| = 80$

$3 - x = 80$ or $3 - x = -80$

$x = -77 \quad x = 83$

Q. **Find area of quadrilateral A(1, 1), B(3, 4), C(5, −2) and D(4, −7) in order are the vertices of a quadrilateral.**

Sol. Area $= \frac{1}{2}\left\|\begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} + \begin{vmatrix} 3 & 5 \\ 4 & -2 \end{vmatrix} + \begin{vmatrix} 5 & 4 \\ -2 & -7 \end{vmatrix} + \begin{vmatrix} 4 & 1 \\ -7 & 1 \end{vmatrix}\right\|$

$$= \frac{1}{2}\left|1 + (-26) + (-27) + (11)\right| = \frac{41}{2}\text{unit}$$

Q. **Find coordinate of point P if PA = PB and area of, ΔPAB = 10, if coordinates of A and B are (3, 0) and (7, 0) respectively.**

Sol. Let $P(\alpha, \beta)$ Now

PA = PB

$\Rightarrow PA^2 = PB^2$

$$(\alpha - 3)^2 + (\beta - 0)^2 = (\alpha - 7)^2 + (\beta - 0)^2$$

$$\alpha^2 - 6\alpha + 9 + \beta^2 = \alpha^2 - 14\alpha + 49 + \beta^2$$

$\Rightarrow 8\alpha = 40 \Rightarrow \alpha = 5$ ... (1)

Now area of $\Delta PAB = 10$

$$\frac{1}{2}\left\|\begin{matrix}\alpha & 3\\ \beta & 0\end{matrix}\right| + \left|\begin{matrix}3 & 7\\ 0 & 0\end{matrix}\right| + \left|\begin{matrix}7 & \alpha\\ 0 & \beta\end{matrix}\right\| = 0$$

$|-3\beta + 7\beta| = 20 \Rightarrow 4\beta = \pm 20$

$\Rightarrow \beta = \pm 5$

Hence, P (5, 5) or P (5, −5)

Q. **Find the area of the Δ. If the coordinate of vertices of triangle are** $(at_1^2, 2at_1), (at_2^2, 2at_2), (at_3^2, 2at_3)$

Sol.

$$\text{Area} = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ at_1^2 & at_2^2 & at_3^2 \\ 2at_1 & 2at_2 & 2at_3 \end{vmatrix}$$

$$= \frac{1}{2} \times a \times 2a \times \begin{vmatrix} 1 & 1 & 1 \\ t_1^2 & t_2^2 & t_3^2 \\ t_1 & t_2 & t_3 \end{vmatrix}$$

$$= a^2|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$$

Locus:

Path traced out by point under one or more given conditions.

To find Locus:

(i) Write geometrical condition and convert them in algebraic.

(ii) Eliminate Variables.

(iii) Get relation between h and k.

(iv) To get equation of locus replace h by x and k by y.

Q. **Find locus of curve / point which is equidistant from points (0, 0) and (2, 0)**

Sol. Let point be P(h, k) and A(0, 0) & B(2, 0)

Given PA = PB

$\Rightarrow PA^2 = PB^2$

$(h-0)^2+(k-0)^2=(h-2)^2+(k-0)^2$
$h^2=h^2-4h+4$
$\Rightarrow h=1$
hence, locus is $x=1$

Q. **If A(0, 0), B(2, 0). Find locus of point P such that $\angle APB = 90°$.**

Sol. Let P(h, k) be point then
$\angle APB = 90° \Rightarrow \Delta APB$ is right angled.
$\Rightarrow AP^2+PB^2=AB^2$
$\Rightarrow (h^2+k^2)+(h-2)^2+k^2=2^2+0^2$
$\Rightarrow 2h^2+2k^2-4h=0$
$\Rightarrow h^2+k^2-2h=0$
locus is $x^2+y^2-2x=0$

Q. **If A(0, 0), B(2, 0). find locus of point P such that area (ΔAPB) = 4**

Sol. Let $P(h,k)$ now

ar $(\Delta APB) = 4$

$$\frac{1}{2}\left\|\begin{vmatrix}0 & h\\ 0 & k\end{vmatrix}+\begin{vmatrix}h & 2\\ k & 0\end{vmatrix}+\begin{vmatrix}2 & 0\\ 0 & 0\end{vmatrix}\right\| = 4 \Rightarrow |2k| = 8 \Rightarrow k = \pm 4$$

locus is $y = \pm 4$

Q. **If A and B are variable point on x and y axis such that length (AB) = 4 find:**
(i) Locus of mid point of AB
(ii) Locus of circumcentre of ΔAOB
(iii) Locus of G of ΔAOB
(iv) Find locus of point which divides segment AB internally in the ratio 1: 2,1 from x-axis

Sol. Let $A(\alpha,0)$ & $B(0,\beta)$

$\because AB = 4 \Rightarrow \alpha^2+\beta^2=16$ ... (1)

(i) let mid point of AB is $P(h,k)$

$$h=\frac{\alpha+0}{2}, k=\frac{0+\beta}{2} \Rightarrow \alpha = 2h, \beta = 2k \text{ from } (1) \Rightarrow 4h^2+4k^2=16 \Rightarrow h^2+k^2=4$$

locus $x^2 + y^2 = 4$

(ii) Circumcentre of ΔAOB will be mid point

of $AB\left(\because \angle AOB = 90^\circ\right)$

hence locus $x^2 + y^2 = 4$

(iii) Let G(h, k) then

$$h = \frac{\alpha + 0 + 0}{3},\ k = \frac{0 + \beta + 0}{3}$$

$\Rightarrow \alpha = 3h,\ \beta = 3k$

Now, from (1)

$9h^2 + 9k^2 = 16$

locus $\Rightarrow 9x^2 + 9y^2 = 16$

(iv) Let the point P(h, k)

$$h = \frac{2\alpha}{3},\ k = \frac{\beta}{3}$$

$$\Rightarrow \alpha = \frac{3h}{2}, \beta = 3k$$

From (1)

$$\frac{9h^2}{4} + 9k^2 = 16$$

$9h^2 + 36k^2 = 64$

locus $\Rightarrow 9x^2 + 36y^2 = 64$

Q. **A(1,2) is a fixed point. A variable point B lies on a curve whose equation is $x^2 + y^2 = 4$. Find the locus of the mid point of AB.**

Sol. Let mid point of AB is P(h, k) and $B(\alpha, \beta)$ now

$$\frac{1+\alpha}{2} = h, \frac{2+\beta}{2} = k \Rightarrow \alpha = 2h - 1, \beta = 2k - 2 \text{ hence } B(2h - 1, 2k - 2)$$

now B lies on curve $x^2 + y^2 = 4$

hence $(2h - 1)^2 + (2k - 2)^2 = 4$

locus is $(2x - 1)^2 + (2y - 2)^2 = 4$

Parametric Point

A function in which variables are linked by one or more independent variables (called the parameters) is called parametric equation.

Q. **Find equation of curve represented parametrically by $x = \cos\theta$, $y = \sin\theta$.**

Sol. $x^2 + y^2 = \cos^2\theta + \sin^2\theta$

$x^2 + y^2 = 1$ which is equation of required curve.

Q. **Find the equation of curve if $x = 2\cos\theta$, $y = \sin\theta$.**

Sol. $\left(\frac{x}{2}\right)^2 + y^2 = 1 \Rightarrow x^2 + 4y^2 = 4$

which is equation of required curve.

Q. **Find the equation of curve if $x = \sec\theta$, $y = 2\tan\theta$**

Sol. $\sec\theta = x, \tan\theta = y/2$

$x^2 - (y/2)^2 = \sec^2\theta - \tan^2\theta = 1 \Rightarrow 4x^2 - y^2 = 4$ which is eq of required curve.

Q. **Find the equation of curve if $x = at^2$, $y = 2at$**

Sol. $y^2 = 4a^2t^2$

$= 4a^2\left(\frac{x}{a}\right) = 4ax$

$y^2 = 4ax$

which is equation of required curve.

Q. **Find locus of point P such that : $PF_1 + PF_2 = 2a$ and $F_1 \equiv (c, 0)$ and $F_2 \equiv (-c, 0)$**

Sol. Let P(h, k)

given $PF_1 = 2a - PF_2$

$(PF_1)^2 = (2a - PF_2)^2 \Rightarrow PF_1^2 = 4a^2 - 4aPF_2 + PF_2^2$

$\Rightarrow (h-c)^2 + k^2 = 4a^2 - 4a\sqrt{(h+c)^2 + k^2} + (h+c)^2 + k^2$

$\Rightarrow 4a\sqrt{(h+c)^2+k^2} = 4a^2 + 4ch$ on squaring

$\Rightarrow a^2\left((h+c)^2+k^2\right) = a^4 + c^2h^2 + 2a^2ch$

$\Rightarrow a^2h^2 + a^2c^2 + 2hca^2 + a^2k^2 = a^4 + c^2h^2 + 2a^2ch$

$\Rightarrow \left(a^2-c^2\right)h^2 + a^2k^2 = a^2\left(a^2-c^2\right) \Rightarrow \frac{h^2}{a^2} + \frac{k^2}{a^2-c^2} = 1$

locus $\frac{x^2}{a^2} + \frac{y^2}{a^2-c^2} = 1$

Q. **For points P ≡ (x_1, y_1) and Q ≡ (x_2, y_2) of the co-ordinate plane, a new distance d(P,Q) is defined $d(P,Q) = |x_1 - x_2| + |y_1 - y_2|$. Let O ≡ (0,0) and A ≡ (3,2). Prove that the set of points in the first quadrant which are equidistant (with respect to the new distance) from O and A consists of the union of a line segment of finite length and an infinite ray. sketch this set in a labelled diagram.**

Sol. Let the point be P(h, k) (h, k ≥ 0)

Now, PA = PB

$|h-0| + |k-0| = |h-3| + |k-2|$

$h + k = |h-3| + |k-2|$

locus $x + y = |x-3| + |y-2|$

If $x \geq 3$ & $y \geq 2$ then

$x + y = x - 3 + y - 2 \Rightarrow 0 = -5$

If $x \geq 3$ & $0 \leq y < 2$ then

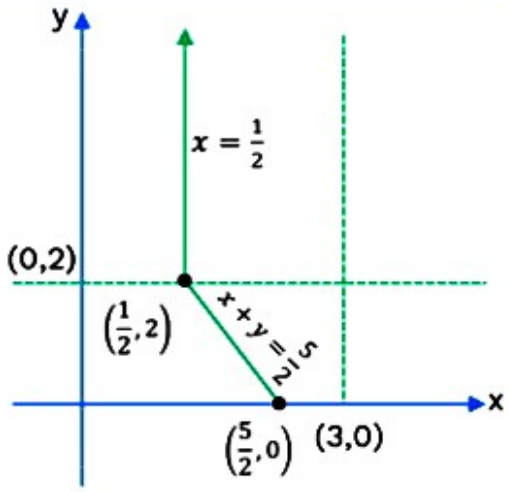
$x + y = x - 3 + 2 - y \Rightarrow y = \frac{-1}{2}$

If $0 \leq x < 3$ & $y \geq 2$ then

$x + y = 3 - x + y - 2 \Rightarrow x = \frac{1}{2}$ (ray)

If $0 \leq x < 3$ & $0 \leq y < 2$

$x + y = 3 - x + 2 - y \Rightarrow x + y = \frac{5}{2}$



Q. **Find locus of point P such that |PA – PB| = 2a and Coordinates of A,B are (c, 0) and (–c, 0).**

Sol. Let Point P(h, k)

PA – PB = ± 2a

PA = ± 2a + PB

$PA^2 = 4a^2 \pm 4aPB + PB^2$

$$(h-c)^2 + k^2 = 4a^2 \pm 4a\sqrt{(h+c)^2 + k^2} + (h+c)^2 + k^2$$

$$\mp 4a\sqrt{(h+c)^2 + k^2} = 4a^2 + 4hc$$

on Squaring

$a^2\{(h + c)^2 + k^2\} = a^4 + 2a^2 hc + h^2c^2$

$a^2h^2 + 2a^2hc + a^2c^2 + a^2k^2 = a^4 + 2a^2hc + h^2c^2$

$\Rightarrow (c^2 - a^2)h^2 - a^2k^2 = a^2(c^2 - a^2)$

$$\Rightarrow \frac{h^2}{a^2} = \frac{k^2}{c^2 - a^2} = 1$$

locus $\dfrac{x^2}{a^2} - \dfrac{y^2}{c^2 - a^2} = 1$

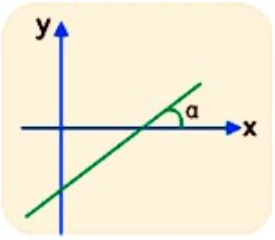
Straight Line

Locus of point such that if any two point of this locus are joined they define a unique direction.

Inclination of Line

The angle of inclination of line is measure of angle between x-axis and line measured in anti-clockwise direction.

$\alpha \in [0,\pi)$



Slope/Gradient (m)

Trignometrical tangent of inclination is said to be slope of line.

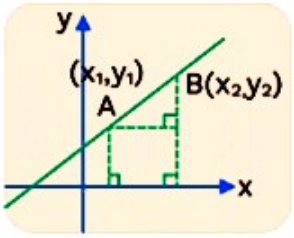
$m = \tan\alpha;\ \alpha \neq \pi/2$

Slope of line Joining two points

If a line passes through two points $A(x_1, y_1)$ and $B(x_2, y_2)$ then its slope will be

$m = \tan\alpha$

$$= \frac{y_2 - y_1}{x_2 - x_1}$$



Q. **Find slope of joining point (1, 1) and (100, 100)**

Sol. $m = \frac{100-1}{100-1} = \frac{99}{99} = 1$

Q. **Find slope of joining points (1, 9) and (7, 0)**

Sol. $m = \frac{9-0}{1-7} = \frac{-9}{6}$

Equation of line in Various Forms :

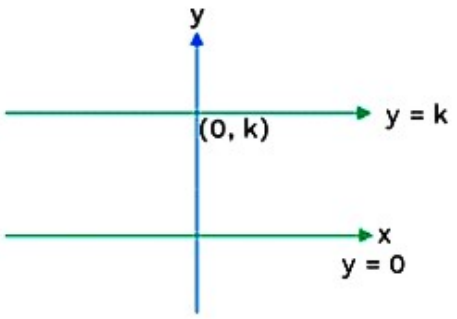
General form :

Equation of any line in plane can be written as

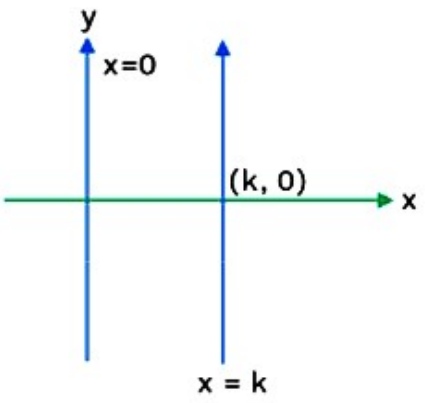
$ax + by = c$

Different forms of a line

(i) Equation of line parallel to x-axis



(ii) Equation of a line parallel to y–axis



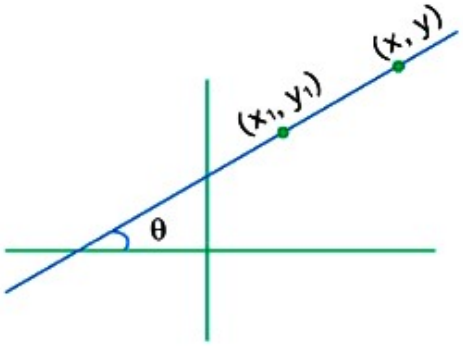
(iii) Point slope form of a line:

The equation of the line passes through the point (x_1, y_1) and has the slope m is

Given slope, $m = \tan\theta$

also $\frac{y-y_1}{x-x_1} = m$

or $y - y_1 = m(x - x_1)$



Q. **Find equation of line having slope 2 and passing through point (1, 3).**

Sol. Equation

$(y - 3) = 2(x - 1)$

$y = 2x + 1$

Q. **Find equation of line having slope $\sqrt{3}$ and passing through point (1, 7).**

Sol. Equation is

$$(y - 7) = \sqrt{3}(x - 1)$$

$$y = \sqrt{3}x + (7 - \sqrt{3})$$

Q. **Line passing though (1, 0) and (2, 1) is rotated about point (1, 0) by an angle 15° in clockwise direction. Find equation of line.**

Sol. Slope of previous line

$$m_1 = \frac{1-0}{2-1} = 1 = \tan 45^\circ$$

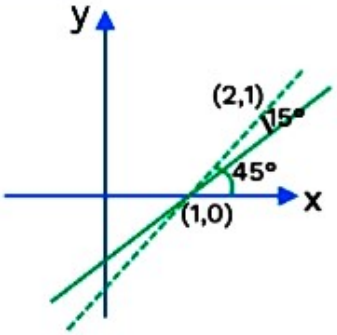
Slope of new line

$m_2 = \tan(45^\circ - 15^\circ)$

$$= \frac{1}{\sqrt{3}}$$

equation $(y - 0) = \frac{1}{\sqrt{3}}(x - 1)$

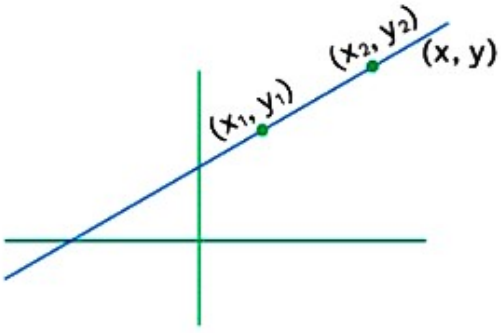
$$y = \frac{1}{\sqrt{3}}x - \frac{1}{\sqrt{3}}$$



Two point form :

The equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$



Proof : Some slope of a line gives

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

So $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Q. **Find equation of line joining (1, 1), (3, 4)**

Sol. Equation $(y-1) = \frac{4-1}{3-1}(x-1)$

$y - 1 = \frac{3}{2}(x-1) \Rightarrow 2y - 2 = 3x - 3$

$\Rightarrow 3x - 2y = 1$

Determinant Form :

Equation of line passing through points $A(x_1, y_1)$ and $B(x_2, y_2)$ will be

$$\begin{vmatrix} 1 & 1 & 1 \\ x & x_1 & x_2 \\ y & y_1 & y_2 \end{vmatrix} = 0$$

Intercept slope form :

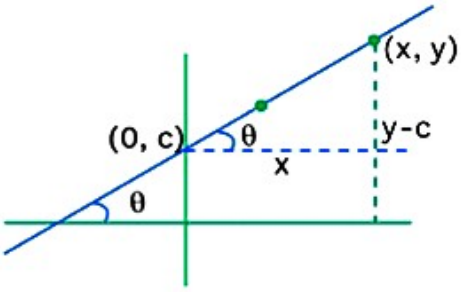
The equation of a line with slope m that makes an intercept c on the y-axis is $y = mx + c$

$I^{st}: \frac{y-c}{x} = m \Rightarrow y = mx + c$

$II^{nd}: \tan\theta = m = \frac{y-c}{x}$

$y = mx + c$

If the line passes through the origin then equation of line is $y = mx$



Q. **Find slope, x – intercept, y – intercept of lines**
(i) 2y = 3x + 7
(ii) 2x + 7y – 3 = 0
(iii) line joining points (1, 1), (9, 3)

Sol. **(i)** $y = \frac{3}{2}x + \frac{7}{2}$

slope $= 3/2$, y – intercept $= 7/2$

put $y = 0 \Rightarrow$ x – intercept $= -7/3$

(ii) $7y = -2x + 3$

$y = \frac{-2}{7}x + \frac{3}{7}$

Slope $= -2/7$, y – intercept $= 3/7$

put $y = 0 \Rightarrow x$ – intercept $= 3/2$

(iii) equation $(y-1) = \frac{3-1}{9-1}(x-1)$

$y - 1 = \frac{1}{4}x - \frac{1}{4} \Rightarrow y = \frac{1}{4}x + \frac{3}{4}$

Slope $= 1/4, y$ – intercept $= 3/4$

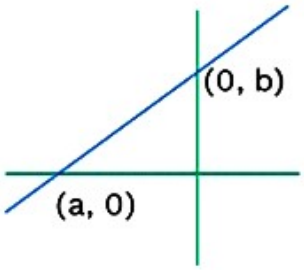
put $y = 0 \Rightarrow x$ – intercept $= -3$

Double Intercept form :

Intercept of line on coordinate axis are points where line intersects x–axis and y–axis is called x intercept and y–intercept.

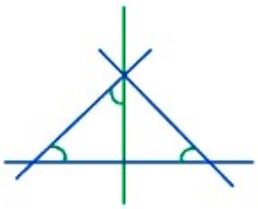
a and b can be positive or negative

The equation of a line which cuts – off intersects a and b respectively, from the x axis and y – axis is $\frac{x}{a} + \frac{y}{b} = 1$



Proof : $y - 0 = \frac{b-0}{0-a}(x-a)$

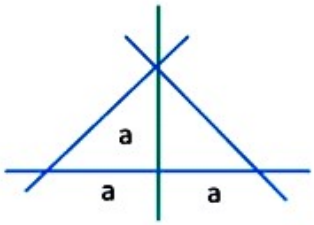
$\frac{y}{b} = -\frac{x}{a} + 1 \Rightarrow \frac{x}{a} + \frac{y}{b} = 1$



If line is equally inclined with coordinate axes then $m = \pm 1$

If line cutting off equal non–zero distance from origin then $m = \pm 1$

$0.x + 0.y + c = 0$ represent a straight line with x and y intercept both infinity $\Rightarrow$ straight line approaches at infinity.



Q. **Find equation of straight line through (1, 2) and if its x – intercept is twice the y – intercept.**

Sol. Let equation $\frac{x}{2a} + \frac{y}{a} = 1$

it passes through (1,2) hence

$\frac{1}{2a} + \frac{2}{a} = 1 \Rightarrow \frac{5}{2a} = 1 \Rightarrow a = 5/2$

$\Rightarrow$ eq $\frac{x}{5} + \frac{2y}{5} = 1 \Rightarrow x + 2y = 5$

Q. **Find equation of line passing through (2, 3) and having intercept on y – axis twice its intercept on x-axis.**

Sol. Let eq^n $\frac{x}{a}+\frac{y}{2a}=1$

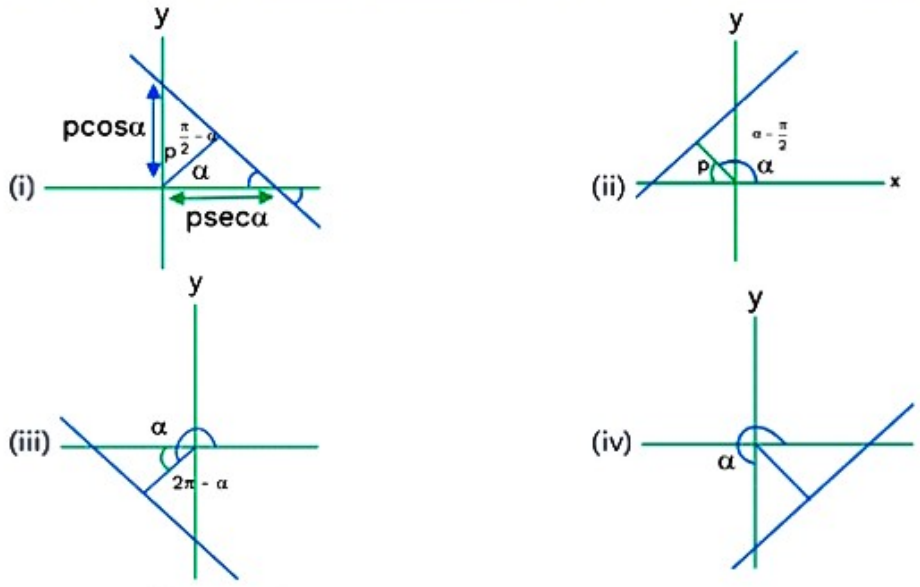
It passes through (2, 3) hence

$\frac{2}{a}+\frac{3}{2a}=1 \Rightarrow \frac{7}{2a}=1 \Rightarrow a = 7/2$

$\Rightarrow$ eq^n $\frac{2x}{7}+\frac{y}{7}=1 \Rightarrow 2x + y = 7$

Normal form (Perpendicular form) :

The equation of the straight line upon which the length of the perpendicular from the origin is p and this perpendicular makes an angle α with the positive direction of the x – axis is $x\cos\alpha + y\sin\alpha = p$



(i) $\frac{x}{p\sec\alpha}+\frac{y}{p\cos\alpha}=1$, $x\cos\alpha + y\sin\alpha = p$

(ii) $\cos(\pi-\alpha)=\frac{p}{a} \Rightarrow a=\frac{p}{-\cos\alpha} \Rightarrow -a=\frac{p}{\cos\alpha}$

Similarly, $b=\frac{p}{\sin\alpha}$

$x\cos\alpha + y\sin\alpha = p$

Q. **If equation of line $3x - 4y + 5 = 0$ convert line in**
(i) Slope intercept form
(ii) Double Intercept form
(iii) Normal form

Sol. **(i)** $4y = 3x + 5$

$$y = \frac{3}{4}x + \frac{5}{4};\ m = \frac{3}{4}, c = \frac{5}{4}$$

(ii) $3x - 4y = -5$

$$\frac{x}{(-5/3)} + \frac{y}{(5/4)} = 1$$

$$a = \frac{-5}{3}, b = \frac{5}{4}$$

(iii) $3x - 4y = -5$

divide by $\sqrt{3^2 + 4^2}$

$$\frac{3}{5}x + y\left(-\frac{4}{5}\right) = -1$$

$$\Rightarrow x\left(\frac{-3}{5}\right) + y\left(\frac{4}{5}\right) = 1$$

$$\cos\alpha = \frac{-3}{5}, \sin\alpha = \frac{4}{5}, p = 1$$

Note :
(i) Line having equal intercept $\Rightarrow m = -1$
(ii) Line equally inclined with co-ordinate axes $\Rightarrow m = \pm 1$

Q. **Find equation of medians of ΔABC where coordinate of vertices are (0, 0), (6, 0), (3, 8).**

Sol. Let A(0,0), B(6,0) & C(3,8)

then mid point of BC $\Rightarrow D\left(\frac{9}{2}, 4\right)$

mid point of AC $\Rightarrow E\left(\frac{3}{2}, 4\right)$

mid point of AB $\Rightarrow F(3,0)$

equation of median

$$AD \Rightarrow (y - 0) = \frac{4 - 0}{\frac{9}{2} - 0}(x - 0) \Rightarrow 9y = 8x$$

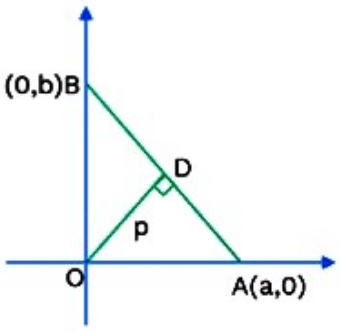
$BE \Rightarrow (y-0) = \dfrac{4-0}{\frac{3}{2}-6}(x-6)$

$y = \dfrac{-8}{9}(x-6) \Rightarrow 8x + 9y = 48$

$CF \Rightarrow x = 3$ ($\because$ slope is not defined)

Q. **If p is perpendicular distance from origin upon line whose intercept on co- ordinate axis are a & b prove that $\dfrac{1}{a^2}+\dfrac{1}{b^2}=\dfrac{1}{p^2}$**

Sol.



$AB = \sqrt{a^2+b^2}$

Now ar(ΔOAB)

$\dfrac{1}{2}.OA.OB = \dfrac{1}{2}.AB.OD$

$ab = \sqrt{a^2+b^2}.p$

on Squaring

$a^2b^2 = (a^2+b^2)p^2$

$\Rightarrow \dfrac{1}{p^2} = \dfrac{a^2+b^2}{a^2b^2} \Rightarrow \dfrac{1}{p^2} = \dfrac{1}{a^2}+\dfrac{1}{b^2}$

Q. **Find locus of mid point of AB where A is x-intercept and B is y-intercept of a variable line always passing through point (2, 3)**

Sol.

Let $A(a,0)$ and $B(0,b)$

also mid point of AB is $P(h,k)$

$\Rightarrow h = \dfrac{a+0}{2}, k = \dfrac{0+b}{2} \Rightarrow a = 2h, b = 2k$

now equation AB $\Rightarrow \dfrac{x}{a}+\dfrac{y}{b} = 1$

$\Rightarrow \dfrac{x}{2h}+\dfrac{y}{2k} = 1$

$\because$ it passes through (2,3) hence

$\dfrac{2}{2h}+\dfrac{3}{2k} = 1$

$\Rightarrow$ locus $\dfrac{1}{x}+\dfrac{3}{2y} = 1 \Rightarrow 2y + 3x = 2xy$

Q. **The region represented by $|x - y| \le 2$ and $|x + y| \le 2$ is bounded by a :**

(A) Square of side length $2\sqrt{2}$ units

(B) rhombus of side length 2 units

(C) square of are 16 sq units

(D) rhombus of area $8\sqrt{2}$ sq. units

Sol. $|x - y| \le 2 \Rightarrow -2 \le x - y \le 2$

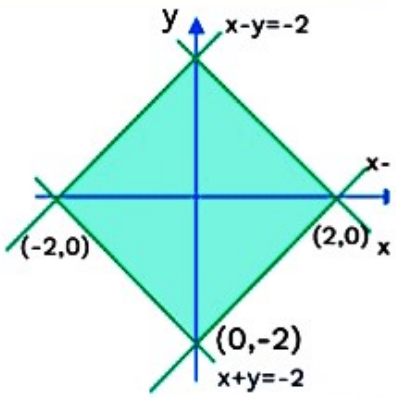
$|x + y| \le 2 \Rightarrow -2 \le x + y \le 2$

Clearly shaded region is an square for which

side $= \sqrt{2^2 + 2^2} = 2\sqrt{2}$

area $= \left(2\sqrt{2}\right)^2$

$= 8$ sq. units



Q. **Find number of lines passing through (2, 4) and forming a triangle of area 16 units with the co-ordinate axes.**

Sol. Let line have equation

$(y - 4) = m(x - 2) \Rightarrow y = mx + 4 - 2m$ ... (1)

put $x = 0 \Rightarrow y = 4 - 2m$

$y = 0 \Rightarrow x = \dfrac{2m - 4}{m}$

line meets co-ordinate axis at $A\left(\dfrac{2m - 4}{m}, 0\right)$

and $B(0, 4 - 2m)$

now area of triangle made by line with co-ordinate axes

$\dfrac{1}{2} OA \cdot OB = 16 \Rightarrow OA.\ OB = 32$

$\left|\dfrac{2m - 4}{m}\right| \times |4 - 2m| = 32$

$(2m - 4)^2 = 32\,|m|$

$\Rightarrow 4(m^2 - 4m + 4) = 32\,|m|$

If $m \ge 0 \Rightarrow m^2 - 4m + 4 = 8m$

$m^2 - 12m + 4 = 0$

$D > 0 \Rightarrow$ two real and distinct values of m

If $m < 0 \Rightarrow m^2 - 4m + 4 = -8m$

$m^2 + 4m + 4 = 0$

$(m + 2)^2 = 0 \Rightarrow m = -2$

$\therefore$ three values of m are possible hence number of such lines = 3

Q. **Find equation of line**

(i) cuts off intercept 4 on x – axis and passing through (2, -3)

(ii) cuts of equal intercept on co-ordinate axes and passes through (2, 5)

(iii) Makes an angle 135° with positive x-axis and cuts y – axis at a distance 8 from the origin

(iv) Passing through (4, 1) and making a triangle with the axes in the first quadrant whose area is 8.

Sol. **(i)** Let equation $\frac{x}{4} + \frac{y}{b} = 1$

Now it passes through (2,–3) hence

$\frac{2}{4} - \frac{3}{b} = 1 \Rightarrow \frac{-1}{2} = \frac{3}{b} \Rightarrow b = -6$

$\therefore$ equation $\frac{x}{4} - \frac{y}{6} = 1$

(ii) Let line $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow x + y = a$

it passes through $(2, 5) \Rightarrow 2 + 5 = a \Rightarrow a = 7$

$\therefore$ equation $x + y = 7$

(iii) Slope $m = \tan(135°) = -1$

or $m = \tan(-135°) = 1$

also $c = \pm 8$

hence, equation is $y = mx + c$

$y = \pm x \pm 8$

(iv) let line $(y - 1) = m(x - 4)$

$y = mx + 1 - 4m$ ... (1)

meets x – axis at $A\left(\frac{4m-1}{m}, 0\right)$

y – axis at B $(0, 1 - 4m)$

area $(\Delta OAB) = 8$

$\frac{1}{2} OA \cdot OB = 8 \Rightarrow OA \cdot OB = 16$

$\left(\frac{4m-1}{m}\right).(1-4m) = 16$

$\Rightarrow (4m-1)^2 = -16m$

($\because$ triangle made is in I quadrant)

$\Rightarrow 16m^2 - 8m + 1 = -16m$

$\Rightarrow 16m^2 + 8m + 1 = 0$

$\Rightarrow (4m+1)^2 = 0 \Rightarrow m = -\frac{1}{4}$

hence, equation (from(1))

$y = -\frac{1}{4}x + 1 - (-1)$

$\Rightarrow 4y = -x + 8$

$x + 4y = 8$

Q. **The number of points, having both coordinates as integers, that lie in the interior of the triangle with vertices (0,0) , (0, 41) and (41, 0)**

(A) 780 **(B) 901** **(C) 861** **(D) 820**

Sol. Equation of sides of Δ are

$x = 0, y = 0, x + y = 41$

If $P(\alpha, \beta)$ is point inside the Δ then

$\alpha > 0, \beta > 0, \alpha + \beta < 41$

If $\alpha = 1 \Rightarrow \beta = 1,2,3, \ldots, 39 \rightarrow 39$ points

If $\alpha = 2 \Rightarrow \beta = 1,2,3, \ldots, 38 \rightarrow 38$ points

$\vdots$

$\alpha = 39 \Rightarrow \beta \rightarrow 1$ point

$\therefore$ total points

$= 1 + 2 + 3 + \ldots + 39 = 39.\frac{40}{2} = 780$

Q. **A straight line L with negative slope passes through the point (8, 2) and cuts the positive coordinate axes at point P and Q. Find the absolute minimum value of OP + OQ. As L varies where O is the origin.**

Sol. Let equation of line

$(y - 2) = m(x - 8)$; $m < 0$

Now $P\left(\frac{8m-2}{m}, 0\right)$, $Q(0, 2-8m)$

Now, $OP + OQ = \frac{8m-2}{m} + 2 - 8m$

$= 8 - \frac{2}{m} + 2 - 8m$

$$= 10 + 2\left\{\frac{1}{(-m)} + 4(-m)\right\} \quad \ldots (1)$$

by AM ≥ GM

$$\frac{1}{(-m)} + 4(-m) \geq 2\sqrt{\frac{1}{(-m)} \times 4(-m)} = 4$$

from (1)

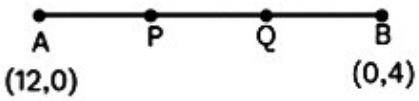
$(OP + OQ)_{min} = 10 + 2(4) = 18$

Q. **Find equation of the two lines which join origin and points of trisection of the portion of line x + 3y – 12 = 0 intercepted between co-ordinate axes.**

Sol. Given line meets coordinate axis at A(12, 0), B(0, 4)

Now, let P, Q are the points which trisects the line segment AB

$$P\left(\frac{1\times 0 + 2\times 12}{1+2}, \frac{1\times 4 + 2\times 0}{1+2}\right) \equiv P\left(8, \frac{4}{3}\right)$$



$$\text{also } Q\left(\frac{8+0}{2}, \frac{\frac{4}{3}+4}{2}\right) \equiv Q\left(4, \frac{8}{3}\right)$$

$$\text{equation of OP} \Rightarrow y = \frac{4/3}{8}x \Rightarrow 6y = x$$

$$\text{equation of OQ} \Rightarrow y = \frac{\frac{8}{3}x}{4} \Rightarrow 3y = 2x$$

For following results a, b and c are the length of sides of triangle and vertex are considered as $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

Line in Determinant form :

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Equation of Median through $A(x_1, y_1)$ in ΔABC

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Equation of Internal angle bisector :

$$b\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + c\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Equation of external angle bisector :

$$b\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - c\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Equation of the altitude :

$$b\cos C\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + c\cos B\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Angle between two line segments :

The angle θ between the line segments having slopes m_1 and m_2 is given by $\tan\theta = \pm\dfrac{m_2 - m_1}{1 + m_1 m_2}$

$m_1 = \tan\theta_1$

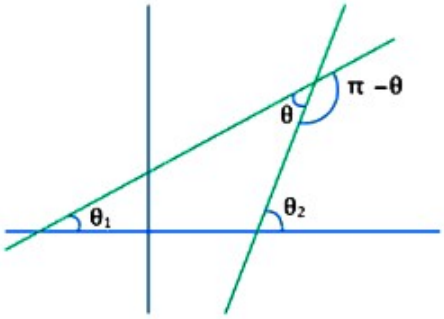
$m_2 = \tan\theta_2$

then $\theta_2 = \theta_1 + \theta$

$\theta = \theta_2 - \theta_1$

$\tan\theta = \tan(\theta_2 - \theta_1)$

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$



But $\pi - \theta$ is also an obtuse angle

$$\tan(\pi - \theta) = -\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

or $$\tan\theta = -\left(\frac{m_2 - m_1}{1 + m_1 m_2}\right)$$

So $\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$

Where θ is the acute angle

Obtuse angle θ is given by $\tan\theta = -\left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$

Parallel : if $\theta = 0$

$$\frac{m_2 - m_1}{1 + m_1 m_2} = 0 \text{ if } m_1 = m_2$$

Perpendicular : if $\theta = 90°$

$$m_1 m_2 = -1$$

If $m_1 m_2 = 1$ then if AB forms α angle with x axis then CD will form $90-\alpha$ with x axis.

Equation of the parallel to ax + by + c = 0 is

$ax + by + \lambda = 0$

Equation of line perpendicular to ax + by + c = 0 is

$bx - ay + \lambda = 0$

Inclination of lines which are complementary

$m_1 m_2 = 1$

Q. **Find equation of line parallel and perpendicular to y = 3 and passing through (2, 7)**

Sol. Let line parallel is $y = \lambda$
$\because$ it passes through (2,7) hence,
$\Rightarrow y = 7$
let line $\perp$ is $x = \mu$
$\because$ it passes through (2,7) hence,
$\Rightarrow x = 2$

Q. **Find equation of line parallel and $\perp$ to x = 1 and passing through (–9, –3).**

Sol. Let line parallel is $x = \lambda$
$\because$ it passes through (-9,-3) hence $\Rightarrow x = -9$

let line $\perp$ is $y = \mu$

$\because$ it passes through (-9,–3) hence $\Rightarrow y = -3$

Q. **Find equation of line parallel and perpendicular to $2x + 3y = 7$ and passing through (2,-3)**

Sol. Let line parallel is $2x + 3y = \lambda$

$\because$ it passes through (2,-3) hence

$2(2) + 3(-3) = \lambda \Rightarrow \lambda = -5$

$\Rightarrow$ equation is $2x + 3y + 5 = 0$

now let line perpendicular is : $3x - 2y = \mu$

$\because$ it passes through (2, –3) hence

$3(2) - 2(-3) = \mu \Rightarrow \mu = 12$

$\Rightarrow$ equation is $3x - 2y = 12$

Q. **If the straight line, $2x - 3y + 17 = 0$ is perpendicular to the line passing through the points (7, 17) and (15, β) the β equals**

(A) $\frac{35}{3}$ **(B)** -5 **(C)** $-\frac{35}{3}$ **(D) 5**

Sol. **(D)**

Line $\perp$ to $2x - 3y + 17 = 0$ is

$3x + 2y = \lambda$ now this passes through (7, 17)

$3(7) + 2(17) = \lambda \Rightarrow \lambda = 55$

$\Rightarrow$ equation is : $3x + 2y = 55$

also it passes through $(15, \beta)$ hence $3(15) + 2(\beta) = 55 \Rightarrow \beta = 5$

Q. **Line $2x + 3y = 7$ intersects co-ordinate axes in A and B. Find perpendicular bisector of AB**

Sol. $2x + 3y = 7 \Rightarrow \frac{x}{7/2} + \frac{y}{7/3} = 1$

hence $A\left(\frac{7}{2}, 0\right)$ and $B\left(0, \frac{7}{3}\right)$

now ⊥ bisector of AB will be passing through mid point of AB : $M\left(\frac{7}{4},\frac{7}{6}\right)$ and ⊥ to given line let Line is $3x - 2y = \lambda$

now $3\left(\frac{7}{4}\right) - 2\left(\frac{7}{6}\right) = \lambda \Rightarrow \frac{63-28}{12} = \lambda$

$\lambda = 35/12$ equation is $\Rightarrow 3x - 2y = \frac{35}{12} \Rightarrow 36x - 24x = 35$

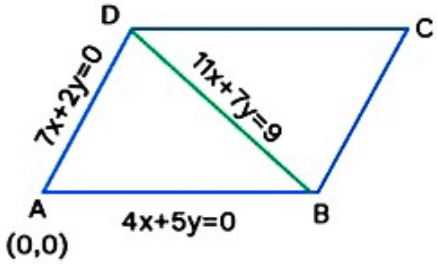
Q. **Two consecutive sides of a parallelogram are 4x + 5y = 0 and 7x + 2y = 0. If the equation to one diagonal is 11x + 7y= 9. find the equation to the other diagonal.**

Sol. Clearly A(0, 0)
Now on solving
11x + 7y = 9
and 4x + 5y = 0

$\Rightarrow 11x + 7\left(\frac{-4x}{5}\right) = 9$

$\Rightarrow \frac{27x}{5} = 9 \Rightarrow x = 5/3, y = -4/3$

$\Rightarrow B\left(\frac{5}{3},\frac{-4}{3}\right)$



on solving 11x + 7y = 9 & 7x + 2y = 0

$\Rightarrow 11x + 7\left(-\frac{7x}{2}\right) = 9 \Rightarrow -\frac{27x}{2} = 9$

$\Rightarrow x = -2/3, y = 7/3 \Rightarrow D\left(-\frac{2}{3},\frac{7}{3}\right)$

now $C\left(\frac{5}{3}-\frac{2}{3}-0,-\frac{4}{3}+\frac{7}{3}-0\right) \equiv C(1,1)$

equation of other diagonal AC is y = x

Q. **Two sides of a parallelogram are along the lines x + y = 3 and x − y + 3 = 0, If its diagonals intersect at (2, 4) then one of its vertex is:**
(A) (2, 6) **(B) (2, 1)** **(C) (3, 5)** **(D) (3, 6)**

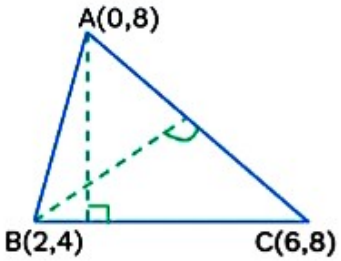
Sol. On solving given lines we get A(0, 3) also P(2, 4) is mid point of AC
$\Rightarrow$ C(4, 5)
Let equation of DC : x + y = λ

on passing it from C we have $x + y = 9$
Now on solving $x - y + 3 = 0$ & $x + y = 9$
D(3, 6)
and B(0 + 4 − 3), (3 + 5 − 6) ≡ B(1, 2)

Q. **A(0, 8), B(2, 4) and C(6, 8). Find equation of altitudes, perpendicular bisector and co-ordinates of orthocenter and circumcentre.**

Sol. $m_{BC} = \frac{4}{4} = 1;$

$m_{AC} = 0; \quad m_{AB} = \frac{4}{-2} = -2$



equation of altitude from A

$(y - 8) = -1(x - 0)$

$x + y = 8$ ... (1)

equation of altitude from B $\Rightarrow x = 2$ ... (2)

equation of altitude from C

$x - 2y + 10 = 0$ ... (3)

on solving (1) and (2)
Orthocentre H(2, 6)

Now for ⊥ bisectors
mid point of BC $\Rightarrow$ D(4,6)
AC $\Rightarrow$ E(3,8)
mid point of AB $\Rightarrow F(1,6)$

equation of ⊥ bisector of BC

$(y - 6) = -1(x - 4) \Rightarrow x + y = 10$ ... (4)

⊥ bisector of AC

$x = 3$ ... (5)

⊥ bisector of AB

$y - 6) = \frac{1}{2}(x - 1) \Rightarrow x - 2y + 11 = 0$ ... (6)

on solving (4) and (5) circumcentre O(3, 7)

To Find Tangent of Interior Angles of Triangle

If m_1, m_2 and m_3 are considered as slope of lines of triangle and if $m_1 > m_2 > m_3$ then

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3}$$

$$\tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$$

Q. **If a ΔABC is formed by the lines $2x + y - 3 = 0$, $x - y + 5 = 0$ and $3x - y + 1 = 0$ then obtain tangents of the interior angles of the triangle.**

Sol. Slopes of sides are $-2, 1, 3$

so let $m_1 = 3, m_2 = 1, m_3 = -2 (\because 3 > 1 > -2)$

now

$$\tan A = \frac{3-1}{1+3(1)} = 1/2, \quad \tan B = \frac{1-(-2)}{1+1(-2)} = -3, \quad \tan C = \frac{(-2)-3}{1+(-2)(3)} = 1$$

Q. **Equation of line passing through (1, 2) making an angle of 45° with the line $2x + 3y = 10$**

Sol. Slope of given line $m_1 = -2/3$

let slope of required line m_2 now

$$\tan 45° = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$1 = \left|\frac{-2/3 - m_2}{1 - 2m_2/3}\right|$$

$$\Rightarrow 1 = \left|\frac{3m_2 + 2}{2m_2 - 3}\right| \Rightarrow |3m_2 + 2| = |2m_2 - 3|$$

$$\Rightarrow 3m_2 + 2 = 2m_2 - 3 \quad \text{or} \quad 3m_2 + 2 = -2m_2 + 3$$

$m_2 = -5$ or $m_2 = 1/5$

equation $(y - 2) = -5(x - 1)$ or $(y - 2) = \frac{1}{5}(x - 1)$

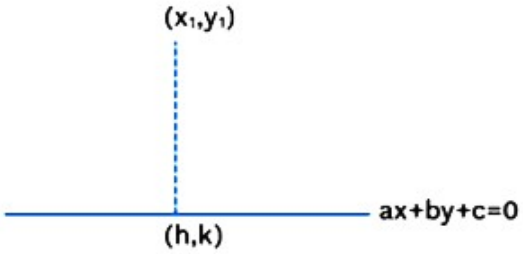
Feet of perpendicular :

Suppose ax + by + c = 0 is the line and (h, k) is the feet of perpendicular of point (x_1, y_1).

The

$ah + bk + c = 0$

$$\left.\begin{array}{ll} \dfrac{y_1 - k}{x_1 - h} = & -\dfrac{a}{b} \\ by_1 - bk = & -ax_1 + ah \\ ax_1 + by_1 = & ah + bk \end{array}\right\} \begin{array}{l} bx - ay = c_1 \\ bx - ay = bx_1 - ay_1 \\ bh - ak = bx_1 - ay_1 \end{array}$$



$$\frac{h - x_1}{a} = \frac{k - y_1}{b}$$

$$\frac{ah - ax_1}{a^2} = \frac{bk - by_1}{b^2} = \frac{ah + bk - ax - by}{a^2 + b^2}$$

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$$

Image of P with respect to L

$$(h, k) = \left(\frac{x_2 + x_1}{2}, \frac{y_1 + y_2}{2}\right)$$

we get

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = -2\frac{(ax_1 + by_1 + c)}{a^2 + b^2}$$

Q. **Find equation of line passing through (-2, -7) making an angle of $\tan^{-1} 3/4$ with the line 4x + 3y = 3**

Sol. Slope of req'd line = m_1
Slope of given line = $m_2 = -4/3$

$$\text{now } \tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$\tan\left(\tan^{-1} 3/4\right) = \left|\frac{m_1 + 4/3}{1 + (-4m_1/3)}\right|$$

$$\Rightarrow \frac{3}{4} = \left|\frac{3m_1 + 4}{4m_1 - 3}\right|$$

$$\Rightarrow 3(4m_1 - 3) = \pm 4(3m_1, +4)$$

$$\Rightarrow 12m_1 - 9 = \pm(12m_1 + 16)$$

$(+\text{sign} \Rightarrow -9 = +16 \Rightarrow m_1 \to \infty$

$(-)\text{sign} \Rightarrow 24m_1 = -7 \Rightarrow m_1 = -7/24$

Now equation $(y + 7) = -\frac{7}{24}(x + 2)$, $x = -2$

Q. **Find reflection of point (1, -2) about the line $x - y + 5 = 0$**

Sol.

$$\frac{x-1}{1} = \frac{y+2}{-1} = -2\left\{\frac{(1)-(-2)+5}{1^2 + (-1)^2}\right\}$$

$$\frac{x-1}{1} = \frac{y+2}{-1} = -2\left(\frac{8}{2}\right)$$

$x = -7, y = 6$

$\therefore$ reflection is $(-7, 6)$

Q. **Equation of perpendicular bisector of the sides AB and AC of a triangle ABC is $x - y + 5 = 0$ and $x + 2y = 0$. If the vertex is A(1, −2). find equation of BC.**

Sol.

Given

$L_1 : x - y + 5 = 0$

$L_2 : x + 2y = 0$

Now L_1 is $\perp$ bisector of AB

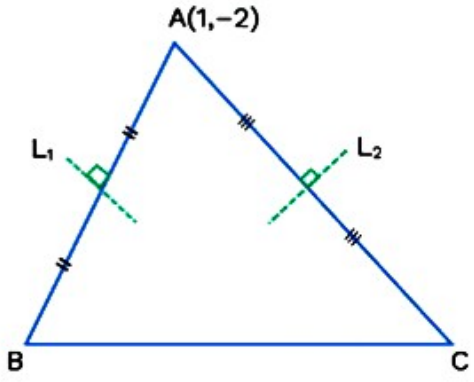
$\Rightarrow$ B is reflection of A in L_1 hence for B

$$\frac{x-1}{1} = \frac{y+2}{-1} = 2\left\{\frac{1-(-2)+5}{1^2 + (-1)^2}\right\}$$

$$\frac{x-1}{1} = \frac{y+2}{-1} = -2\left(\frac{8}{2}\right) \Rightarrow B(-7, 6)$$

Now L_2 is $\perp$ bisector of AC

$\Rightarrow$ C is reflection of A in L_2 hence for C



$$\frac{x-1}{1}=\frac{y+2}{2}=-2\left\{\frac{1+2(-2)}{1^2+2^2}\right\}$$

$$\frac{x-1}{1}=\frac{y+2}{2}=-2\left(\frac{-3}{5}\right)$$

$$\Rightarrow C\left(\frac{11}{5},\frac{2}{5}\right)$$

$$m_{BC}=\frac{6-2/5}{-7-11/5}=\frac{28/5}{-46/5}=-\frac{14}{23}$$

equation of BC $(y-6)=\frac{-14}{23}(x+7)$

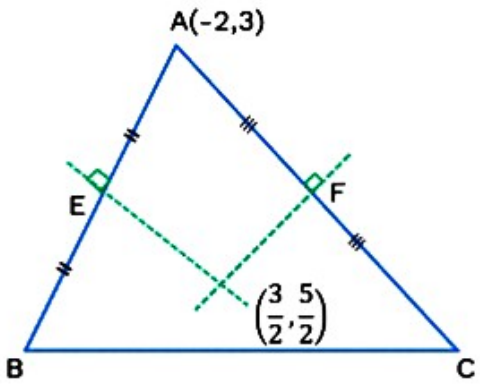
$\Rightarrow 14x+23y=40$

Q. **The equations of perpendiculars of the sides AB and AC of triangle ABC are $x-y-4=0$ and $2x-y-5=0$ respectively. If the vertex A is (–2, 3) and point of intersection of perpendiculars bisectors is (3/2, 5/2), find the equation of medians to the sides AB and AC respectively.**

Sol. Given circumcentre is $O\left(\frac{3}{2},\frac{5}{2}\right)$

also : $L_1 : x-y-4=0$
$L_2 : 2x-y-5=0$
$\perp$ bisector of AB will be parallel to L_1 and passes from O hence equation $L_3 : x-y+1=0$
$\perp$ bisector of AC will be $\parallel$ to L_2 and passes from O hence

eq^n : $2x-y-\frac{1}{2}=0 \Rightarrow L_4 : 4x-2y-1=0$



now E is foot of $\perp$ from A on L_3 hence,

$$\frac{x+2}{1}=\frac{y-3}{-1}=-1\left\{\frac{-2-3+1}{1^2+(-1)^2}\right\}$$

$\Rightarrow$ E(0, 1) hence, B(2, –1)
also F is foot of $\perp$ from A on L_4 hence

$$\frac{x+2}{4}=\frac{y-3}{-2}=-\left\{\frac{4(-2)-2(3)-1}{16+4}\right\}$$

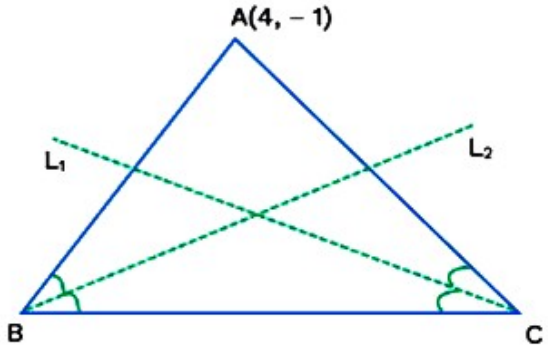
$\Rightarrow$ F(1, 3/2) Hence, C(4, 0)
equation of median of side AB
EC : $x + 4y = 4$
equation of median of side AC

$$FB : (y + 1) = \frac{\frac{3}{2} + 1}{1 - 2}(x - 2)$$

$\Rightarrow 2y + 2 = -5x + 10 \Rightarrow 5x + 2y = 8$

Q. **Find the equation of the sides of triangle having (4, –1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.**

Sol. Let given A(4,-1) and angle bisector of



angle B
$L_1 : x - 1 = 0$
and of angle C is
$L_2 : x - y - 1 = 0$
Clearly of reflection of A in L_1 and L_2 lies on BC hence reflection of A in L_1

$$\frac{x - 4}{1} = \frac{y + 1}{0} = -2\left\{\frac{4 - 1}{1^2}\right\}$$

$\Rightarrow$ A'(–2, – 1)
now reflection of A in L_2

$$\frac{x - 4}{1} = \frac{y + 1}{-1} = -2\left\{\frac{4 + 1 - 1}{1^2 + (-1)^2}\right\}$$

$\Rightarrow$ A"(0, 3)
equation of BC = equation of A'A"

$$(y + 1) = \frac{4}{2}(x + 2) \Rightarrow 2x - y + 3 = 0$$

on solving BC with $L_1 \Rightarrow$ B(1, 5) on solving BC with $L_2 \Rightarrow$ C(–4, –5)

now equation of AB $\Rightarrow (y - 5) = \frac{6}{-3}(x - 1)$

$\Rightarrow 2x + y = 7$

equation of AC $\Rightarrow (y + 5) = \frac{4}{8}(x + 4)$

$\Rightarrow x - 2y = 6$

Q. **Find reflection of point (1, –2) about the line x + 2y = 0**

Sol.

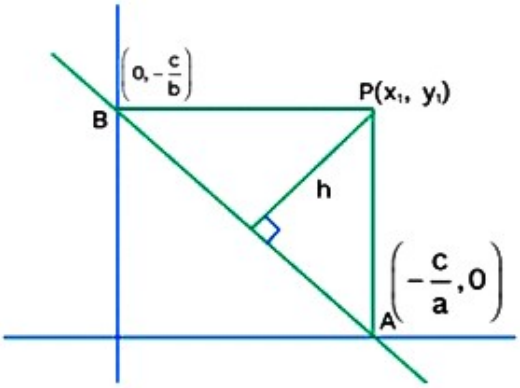
$$\frac{x-1}{1}=\frac{y+2}{2}=-2\left\{\frac{1+2(-2)}{5}\right\}$$

$$\frac{x-1}{1}=\frac{y+2}{2}=\frac{6}{5}$$

$$\Rightarrow x=\frac{11}{5},\ y=\frac{2}{5}$$

Reflection is $\left(\frac{11}{5},\frac{2}{5}\right)$

Length of perpendicular or distance of a point from a line :



Let AB be the line ax + by + c = 0

$$[APB]=\frac{1}{2}\begin{Vmatrix}-\frac{c}{a} & 0 & 1\\ x & y & 1\\ 0 & -\frac{c}{b} & 1\end{Vmatrix}$$

Also, $AB=\sqrt{\frac{c^2}{a^2}+\frac{c^2}{b^2}}$

$$=c\sqrt{\frac{1}{a^2}+\frac{1}{a^2}}$$

$$[APB]=\frac{1}{2}\left|x_1\frac{c}{b}+y_1\frac{c}{a}+\frac{c^2}{ab}\right|=\frac{1}{2}c\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}\,h$$

ax + by + c = 0

$$\left|c\frac{(ax_1+by_1+c)}{ab}\right|=c\,h\sqrt{\frac{a^2+b^2}{a^2b^2}}$$

$$h=\left|\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|$$

Rule-1.

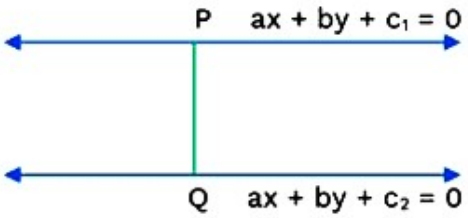
The length of the perpendicular from the origin to the line $ax + by + c = 0$ is

$$\left|\frac{c}{\sqrt{a^2+b^2}}\right|$$

Rule-2.

The distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is given by

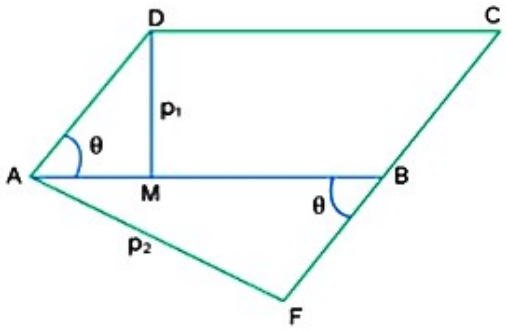
$$\left|\frac{c_1 - c_2}{\sqrt{a^2+b^2}}\right|$$



Rule-3.

The area of a parallelogram whose sides are $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_2x + b_2y + d_2 = 0$ is given by

$$\frac{p_1 \times p_2}{\sin\theta} = \left|\frac{(c_1 - d_1)(c_2 - d_2)}{(a_1b_2 - a_2b_1)}\right|$$



Proof:

The distance between two parallel sides $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + d_1 = 0$ is given by

$$p_1 = \left|\frac{(c_1 - d_1)}{\sqrt{a_1^2 + b_1^2}}\right|$$

The distance between two parallel sides $a_2x + b_2y + c_2 = 0$, $a_2x + b_2y + d_2 = 0$ is given by

$$p_2 = \left|\frac{(c_2 - d_2)}{\sqrt{a_2^2 + b_2^2}}\right|$$

$$\text{Now, } \tan\theta = \left|\frac{\frac{b_1}{a_1} - \frac{b_2}{a_2}}{1 + \frac{b_1}{a_1}\cdot\frac{b_2}{a_2}}\right| = \left|\frac{a_2b_1 - a_1b_2}{a_1a_2 + b_1b_2}\right|$$

$$\Rightarrow \quad \sin\theta = \frac{a_1b_2 - a_2b_1}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}$$

$$\text{Thus, the area of a parallelogram} = \frac{p_1 \times p_2}{\sin\theta}$$

$$= \frac{(c_1 - d_1)(c_2 - d_2)}{(a_1b_2 - a_2b_1)}$$

Rule-4.

If $p_1 = p_2$, the parallelogram becomes a rhombus and its area is given by

$$\frac{(c_1 - d_1)^2}{\left|(a_1b_2 - a_2b_1)\right|\sqrt{\left(\frac{a_1^2 + b_1^2}{a_2^2 + a_2^2}\right)}}.$$

Rule-5.

The area of a parallelogram whose sides are $y = mx + a$, $y = mx + b$, $y = nx + c$ and $y = nx + d$ is given by $\left|\frac{(a-b)(c-d)}{(m-n)}\right|$.

Q. **Find distance between point (1,2) and line $3x - 4y + 1 = 0$**

Sol.

$$P = \left|\frac{3(1) - 4(2) + 1}{\sqrt{3^2 + 4^2}}\right|$$

$$P = \frac{4}{5}$$

Q. **Find Distance between point (0, 0) line $12x - 5y + 7 = 0$**

Sol.

$$P = \left|\frac{12(0) - 5(0) + 7}{\sqrt{12^2 + 5^2}}\right| = \frac{7}{13}$$

Q. **Lines are drawn parallel to the line $4x - 3y + 2 = 0$ at a distance $\frac{3}{5}$ from the origin. Then which one of the following points lies on any of these lines?**

(A) $\left(-\frac{1}{4}, \frac{2}{3}\right)$ **(B)** $\left(\frac{1}{4}, \frac{1}{3}\right)$ **(C)** $\left(\frac{-1}{4}, -\frac{2}{3}\right)$ **(D)** $\left(\frac{1}{4}, -\frac{1}{3}\right)$

Sol. **(A)**

Let line || to it is

$L : 4x - 3y + \lambda = 0$

Now distance of O from L is $\frac{3}{5}$

hence, $\left|\frac{4(0)-3(0)+\lambda}{\sqrt{4^2+3^2}}\right| = \frac{3}{5}$

$\Rightarrow |\lambda| = 3 \Rightarrow \lambda = \pm 3$

hence lines are $4x - 3y + 3 = 0$

or $4x - 3y - 3 = 0$

Q. **The locus of the mid points of the perpendicular drawn from points on the line $x = 2y$ to the line $x = y$ is**

(A) $2x - 3y = 0$ (B) $3x - 2y = 0$ (C) $5x - 7y = 0$ (D) $7x - 5y = 0$

Sol. **(C)**

Let a point on $L_1 : x = 2y$ is $A(2\alpha, \alpha)$

now foot of $\perp$ from A on $L_2 : x - y = 0$

$$\frac{x-2\alpha}{1} = \frac{y-\alpha}{-1} = -1\left\{\frac{2\alpha-\alpha}{1^2+(-1)^2}\right\}$$

$$x - 2\alpha = -(y-\alpha) = -\frac{\alpha}{2}$$

$$\Rightarrow x = 3\alpha/2,\ y = 3\alpha/2 \Rightarrow B(3\alpha/2, 3\alpha/2)$$

mid point of AB is P(h, k)

$$h = \frac{2\alpha + 3\alpha/2}{2}, k = \frac{\alpha + 3\alpha/2}{2}$$

$4h = 7\alpha$ ... (1)

$4k = 5\alpha$ ... (2)

$(1) \div (2)\ \frac{h}{k} = \frac{7}{5} \Rightarrow 5h = 7k$

locus $\Rightarrow 5x = 7y$

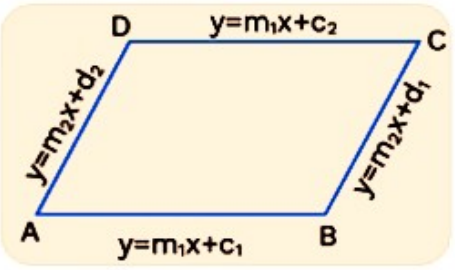
Q. **Find the distance between lines $3x + 4y + 7 = 0$ and $6x + 8y - 17 = 0$**

Sol. $L_1 : 6x + 8y + 14 = 0$

$L_2 : 6x + 8y - 17 = 0$

distance $p = \left|\frac{14-(-17)}{\sqrt{6^2+8^2}}\right| = \frac{31}{10}$

Area of Parallelogram



If we consider lines of the parallelogram as

$y = m_1x + c_1$

$y = m_1x + c_2$

$y = m_2x + d_1$

$y = m_2x + d_2$

then

$$\text{Area} = \left|\frac{(c_1 - c_2)(d_1 - d_2)}{(m_1 - m_2)}\right|$$

Q. **Area of quadrilateral formed by the lines**
$3x - 4y + 10 = 0, 3x - 4y + 20 = 0$
$4x + 3y + 10 = 0, 4x + 3y - 20 = 0$

Sol. Lines are

$L_1 : y = \left(\frac{3}{4}\right)x + \left(\frac{5}{2}\right)$

$L_2 : y = \left(\frac{3}{4}\right)x + 5$

$L_3 : y = \left(\frac{-4}{3}\right)x + \left(\frac{-10}{3}\right)$

$L_4 : y = \left(\frac{-4}{3}\right)x + \left(\frac{20}{3}\right)$

$$\text{Area} = \left|\frac{\left(5 - \frac{5}{2}\right)\left(\frac{20}{3} + \frac{10}{3}\right)}{\frac{3}{4} + \frac{4}{3}}\right| = \frac{\left(\frac{5}{2}\right)(10)}{\frac{25}{12}} = 12 \text{ sq. unit.}$$

Q. **Find the area of quadrilateral formed by the lines**
$3x - 4y + 1 = 0$; $3x - 4y + 2 = 0$
$x - 2y + 3 = 0$; $x - 2y + 7 = 0$

Sol. Sides are $L_1 : y = \left(\frac{3}{4}\right)x + \left(\frac{1}{4}\right)$

$L_2 : y = \left(\frac{3}{4}\right)x + \left(\frac{2}{4}\right)$

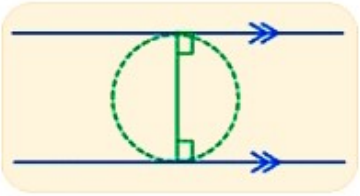
$$L_3 : y = \left(\frac{1}{2}\right)x + \left(\frac{3}{2}\right)$$

$$L_4 : y = \left(\frac{1}{2}\right)x + \left(\frac{7}{2}\right)$$

$$\text{area} = \left|\frac{\left(\frac{2}{4}-\frac{1}{4}\right)\left(\frac{7}{2}-\frac{3}{2}\right)}{\left(\frac{3}{4}-\frac{1}{2}\right)}\right| = \left|\frac{\frac{1}{4}.2}{\frac{1}{4}}\right| = 2 \text{ sq. unit}$$

Note :

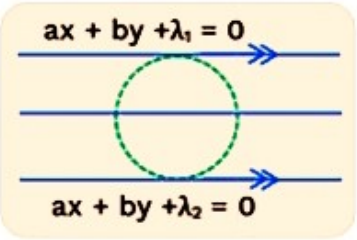
Two parallel lines are tangent to same circle then distance between them is diameter of the circle



Note :

Equation of diameter of tangents is

$$ax + by + \left(\frac{\lambda_1 + \lambda_2}{2}\right) = 0$$



Area of right isosceles Δ in term of perpendicular from vertex

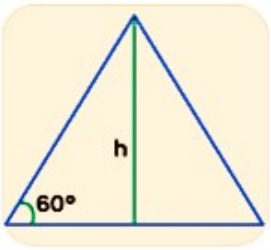
If p is the length of perpendicular from vertex to hypotenuse, then

$\Delta = p^2$

Area of equilateral Δ in terms of median/angle bisector / perpendicular bisector / altitude

If h is the length of median, then

$$\Delta = \frac{h^2}{\sqrt{3}}$$



Q. **Find area of equilateral Δ whose one vertex is (7, 0) and a side lies along line y = x**

Sol. $h = \left|\frac{7-0}{\sqrt{2}}\right| = \frac{7}{\sqrt{2}}$

Area $\Delta = \frac{h^2}{\sqrt{3}} = \frac{49}{2\sqrt{3}}$ sq. unit

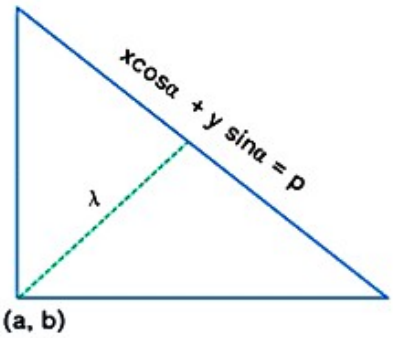
Q. **Two mutually perpendicular lines are drawn passing through point (a, b) and enclosed in an isosceles Δ together with the line x cosα + y sinα = p Find the area of Δ.**

Sol. Given triangle is right angled isosceles Δ hence area Δ $= \lambda^2$ ... (1)

$\lambda = \left|\frac{a\cos\alpha + b\sin\alpha - p}{\sqrt{\cos^2\alpha + \sin^2\alpha}}\right|$

$\lambda = |a\cos\alpha + b\sin\alpha - p|$

area $= \Delta = (a\cos\alpha + b\sin\alpha - p)^2$



Q. **The three lines x + 2y + 3 = 0, x + 2y – 7 = 0 and 2x – y – 4 = 0 form three sides of a square find the equations of the fourth side.**

Sol. Distance between L_1 & L_2

$d = \left|\frac{3+7}{\sqrt{1+4}}\right| = \frac{10}{\sqrt{5}} = 2\sqrt{5}$ unit

let fourth side $L_4 : 2x - y + \lambda = 0$

also distance between L_3 & L_4 will be d

$\left|\frac{\lambda+4}{\sqrt{4+1}}\right| = 2\sqrt{5} \Rightarrow |\lambda + 4| = 10$

$\lambda + 4 = \pm 10$

$\lambda = 6, -14$

hence fourth side: 2x – y + 6 = 0

or 2x -y – 14 = 0

Distance form (Parametric form / Symmetric form):

The equation of the straight line passing through (x_1, y_1) and making an angle θ with positive direction of the x – axis is

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

where r is the distance of the point (x, y) on the line from the point (x_1, y_1)

Proof:

$$\cos\theta = \frac{x - x_1}{r} \quad \text{... (i)}$$

$$x = x_1 + r\cos\theta$$

$$\sin\theta = \frac{y - y_1}{r} \quad \text{... (ii)}$$

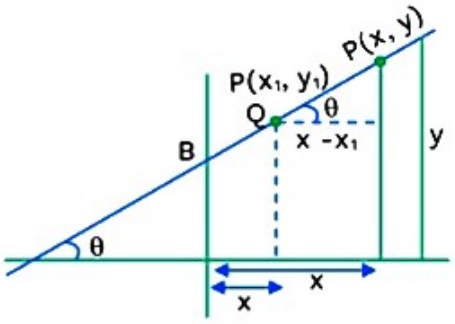
$$y = y_1 + r\sin\theta$$

from (i) and (ii), we get

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

the coordinates of any point on the line at a distance r from the given point (x_1, y_1) is

$(x_1 \pm r\cos\theta, y_1 \pm r\sin\theta)$ + sign of r is above the given point (x_1, y_1) else – sign will be taken.



Q. **In what direction a line through point (1,2) must be drawn so that its intersection point P with the line x + y = 4 may be at a distance of $\frac{\sqrt{6}}{3}$ from A**

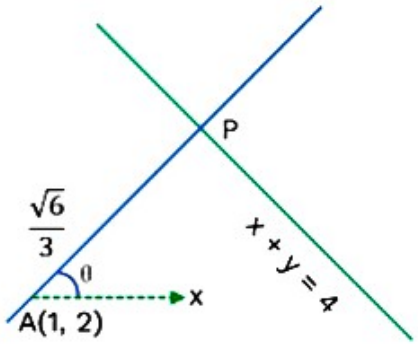
Sol. Let required angle is θ from positive x – axis hence, co-ordinate of P

$$P\left(1 + \frac{\sqrt{6}}{3}\cos\theta, 2 + \frac{\sqrt{6}}{3}\sin\theta\right)$$

Now P lies on x + y = 4

Hence, $1 + \frac{\sqrt{6}}{3}\cos\theta + 2 + \frac{\sqrt{6}}{3}\sin\theta = 4$

$$\frac{\sqrt{6}}{3}(\sin\theta + \cos\theta) = 1 \Rightarrow \sin\theta + \cos\theta = \frac{3}{\sqrt{6}}$$



$$\Rightarrow \frac{1}{\sqrt{2}}(\sin\theta + \cos\theta) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin(\theta + \pi/4) = \sqrt{3}/2$$

$$\theta + \pi/4 = \pi/3,\ \theta + \pi/4 = 2\pi/3 \Rightarrow \theta = \pi/12, 5\pi/12$$

Q. **If A(3, 2), B(7, 4). Find coordinate of C such that ΔABC is equilateral**

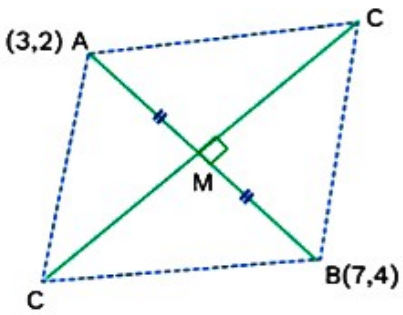
Sol. $AB = \sqrt{4^2 + 2^2} = 2\sqrt{5}$ unit.

hence, length of altitude of ΔABC

$$h = \frac{\sqrt{3}}{2}AB = \sqrt{15} \text{ unit.}$$

$$m_{AB} = \frac{4-2}{7-3} = \frac{1}{2}$$

mid point of AB is M(5, 3) Slope $m_{MC} = -2 = \tan\theta$



$$\Rightarrow \sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{-1}{\sqrt{5}} (\theta \in I)$$

$$\text{or } \sin\theta = -\frac{2}{\sqrt{5}},\ \cos\theta = \frac{1}{\sqrt{5}} (\theta \in I)$$

Now $C(x_1 + h\cos\theta, y_1 + h\sin\theta)$

$$C\left(5 + \sqrt{15}\left(\frac{-1}{\sqrt{5}}\right), 3 + \sqrt{15}\left(\frac{2}{\sqrt{5}}\right)\right) \equiv C(5 - \sqrt{3}, 3 + 2\sqrt{3})$$

$$C\left(5 + \sqrt{15}\left(\frac{1}{\sqrt{5}}\right), 3 + \sqrt{15}\left(\frac{-2}{\sqrt{5}}\right)\right) \equiv C(5 + \sqrt{3}, 3 - 2\sqrt{3})$$

Q. **A line passing through A(−5, −4) meets the line x + 3y + 2 = 0, 2x + y + 4 = 0 and x − y − 5 = 0 at B,C,D. If $\left(\frac{15}{AB}\right)^2 + \left(\frac{10}{AC}\right)^2 = \left(\frac{6}{AD}\right)^2$. Find the equation of line**

Sol. Given lines

$L_1 : x + 3y + 2 = 0$

$L_2 : 2x + y + 4 = 0$

$L_3 : x - y - 5 = 0$

let required line makes angle θ with positive x-axis hence,

$B(-5 + AB\cos\theta, -4 + AB\sin\theta)$

B lies on L_1 hence,

$(-5 + AB\cos\theta) + 3(-4 + AB\sin\theta) + 2 = 0$

$$\frac{15}{AB} = \cos\theta + 3\sin\theta \quad \ldots (1)$$

also $C\left(-5 + AC\cos\theta, -4 + AC\sin\theta\right)$

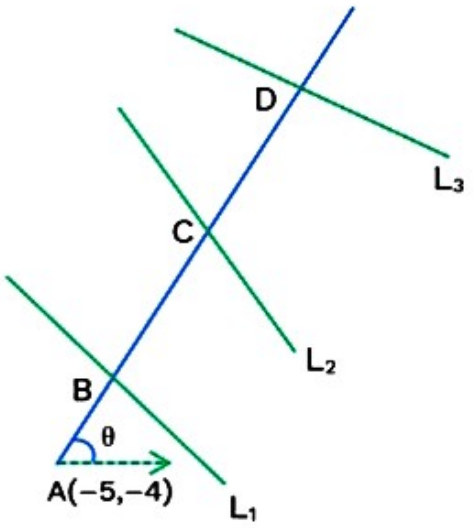
$$2\left(-5 + AC\cos\theta\right) + \left(-4 + AC\sin\theta\right) + 4 = 0$$

$$\Rightarrow \frac{10}{AC} = 2\cos\theta + \sin\theta \quad \ldots (2)$$

and $D\left(-5 + AD\cdot\cos\theta, -4 + AD\cdot\sin\theta\right)$

$$\left(-5 + AD\cos\theta\right) - \left(-4 + AD\sin\theta\right) - 5 = 0$$

$$\Rightarrow \frac{6}{AD} = \cos\theta - \sin\theta \quad \ldots (3)$$



Now, given $\left(\frac{15}{AB}\right)^2 + \left(\frac{10}{AC}\right)^2 = \left(\frac{6}{AD}\right)^2$

from equation (1), (2) & (3)

$(\cos\theta + 3\sin\theta)^2 + (2\cos\theta + \sin\theta)^2 = (\cos\theta - \sin\theta)^2$

$\Rightarrow 4\cos^2\theta + 9\sin^2\theta + 12\sin\theta\cos\theta = 0$ divide by $\cos^2\theta$

$\Rightarrow 4 + 9\tan^2\theta + 12\tan\theta = 0$

$\Rightarrow (3\tan\theta + 2)^2 = 0 \Rightarrow \tan\theta = -2/3$

hence equation of the line

$$\left(y + 4\right) = -\frac{2}{3}\left(x + 5\right) \Rightarrow 2x + 3y + 22 = 0$$

Q. **A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point (1, 5) obtain the equation.**

Sol. Given

$L_1 : 5x - y - 4 = 0$

$L_2 : 3x + 4y - 4 = 0$

let required line makes angle θ with positive x – axis hence,

$C\left(1 + r\cos\theta, 5 + r\sin\theta\right)$

C lies on L_2 hence,

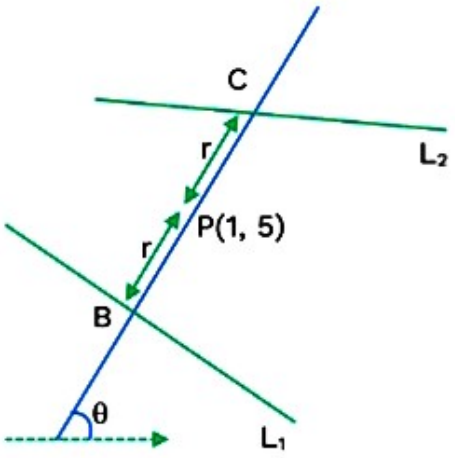
$3\left(1 + r\cos\theta\right) + 4\left(5 + r\sin\theta\right) - 4 = 0$

$\frac{-19}{r} = 3\cos\theta + 4\sin\theta$ ... (1)

Similarly $B(1 - r\cos\theta, 5 - r\sin\theta)$

$5(1 - r\cos\theta) - (5 - r\sin\theta) - 4 = 0$

$\frac{4}{r} = \sin\theta - 5\cos\theta$ ... (2)



(1) ÷ (2) ,

$\frac{-19}{4} = \frac{3\cos\theta + 4\sin\theta}{\sin\theta - 5\cos\theta}$

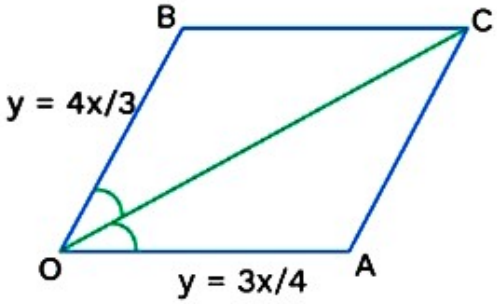
$83\cos\theta = 35\sin\theta \Rightarrow \tan\theta = 83/35$

equation $(y - 5) = \frac{83}{35}(x - 1)$

$\Rightarrow 83x - 35y + 92 = 0$

Q. **Two side of a rhombus lying in 1st quadrant are given by $y = \frac{3x}{4}$ & $y = \frac{4x}{3}$. If the length of longer diagonal OC = 12. Find the equation of other two sides.**

Sol.



$m_{OA} = 3/4$

$m_{OB} = 4/3$ let $m_{OC} = m$

$\because$ OC makes equal angles with OA and OB hence

$$\left|\frac{m - 3/4}{1 + m(3/4)}\right| = \left|\frac{m - \frac{4}{3}}{1 + m\left(\frac{4}{3}\right)}\right|$$

$$\left|\frac{4m - 3}{4 + 3m}\right| = \left|\frac{3m - 4}{4m + 3}\right|$$

$$\left|16m^2 - 9\right| = \left|9m^2 - 16\right|$$

$16m^2 - 9 = 9m^2 - 16$ or $16m^2 - 9 = 16 - 9m^2$

$7m^2 = -7$ or $25m^2 = 25$

$m = \pm 1$

clearly

$m_{OC} = 1 = \tan 45°$

now

$C\left(0+12\cos45^\circ, 0+12\sin45^\circ\right) = C\left(6\sqrt{2}, 6\sqrt{2}\right)$

equation of BC $\Rightarrow \left(y-6\sqrt{2}\right) = \frac{3}{4}\left(x-6\sqrt{2}\right)$

equation of AC $\Rightarrow \left(y-6\sqrt{2}\right) = \frac{4}{3}\left(x-6\sqrt{2}\right)$

Q. **A variable line through origin meets the line $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ at P and Q. On it is taken a point R if $\frac{2}{OR} = \frac{1}{OP} + \frac{1}{OQ}$ then prove that locus of R is also a straight line.**

Sol. Given lines

$L_1 : a_1x + b_1y + c_1 = 0$

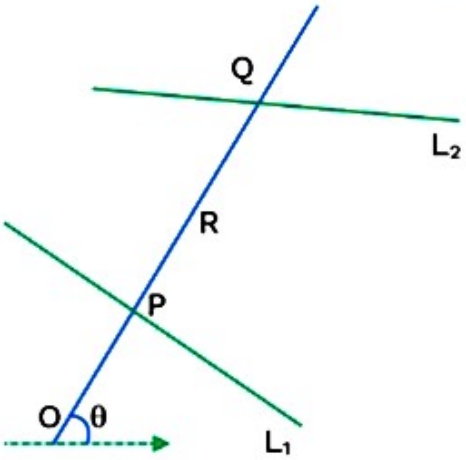
$L_2 : a_2x + b_2y + c_2 = 0$

Let R(h,k)

and $OP = r_1$, $OR = r_2$, $OQ = r_3$

hence $P\left(r_1\cos\theta, r_1\sin\theta\right)$

in putting L_1 we get $\frac{1}{r_1} = \frac{a_1\cos\theta + b_1\sin\theta}{-c_1}$ ... (1)



also $Q\left(r_3\cos\theta, r_3\sin\theta\right)$

in pulting L_2 we get

$$\frac{1}{r_3} = \frac{a_2\cos\theta + b_2\sin\theta}{-c_2}$$

now given $\frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3}$

$$\frac{2}{r_2} = \left\{\frac{a_1\cos\theta + b_1\sin\theta}{-c_1}\right\} + \left\{\frac{a_2\cos\theta + b_2\sin\theta}{-c_2}\right\}$$

$$2 = \frac{a_1\left(r_2\cos\theta\right) + b_1\left(r_2\sin\theta\right)}{-c_1} + \frac{a_2\left(r_2\cos\theta\right) + b_2\left(r_2\sin\theta\right)}{-c_2}$$

$$-2 = \frac{a_1h + b_1k}{c_1} + \frac{a_2h + b_2k}{c_2}$$

$(a_1c_2 + a_2c_1)h + (b_1c_2 + b_2c_1)k + 2c_1c_2 = 0$

locus

$(a_1c_2 + a_2c_1)x + (b_1c_2 + b_2c_1)y + 2c_1c_2 = 0$

$\Rightarrow$ It is a straight line.

Position of a point with respect to a line :

Let the equation of the given line be

$$ax + by + c = 0 \qquad ...(i)$$

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be any two points.

Then R

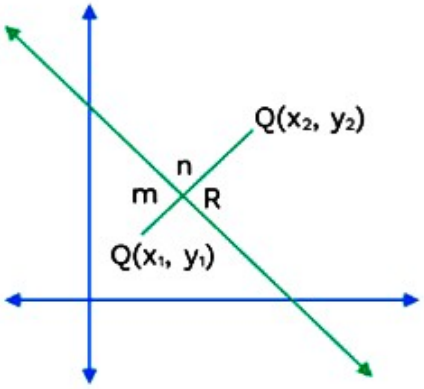
$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$ lies on line

So, $a\left(\frac{mx_2 + nx_1}{m+n}\right) + b\left(\frac{my_2 + ny_1}{m+n}\right) + c = 0$

$a(mx_2 + nx_1) + b(my_2 + ny_1) + c(m + n) = 0$

$m(ax_2 + by_2 + c) + n(ax_1 + by_1 + c) = 0$

$$\frac{m}{n} = -\frac{(ax_1 + by_1 + c)}{(ax_2 + by_2 + c)} > 0$$



If the point R is between the points P and Q i.e., point P and Q are on the opposite side of the given line, then the ratio m : n is positive.

So, we get $\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} < 0$

i.e. $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of the opposite signs.

If P and Q on the same side then $\frac{m}{n}$ is negative

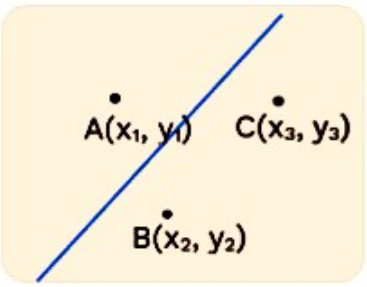
so $\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} > 0$

i.e., $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of the same sign.

Position of Point w.r.t a line

$(ax_2 + by_2 + c)(ax_3 + by_3 + c) > 0$
this means point B and C lies on the same side of line

$(ax_1 + by_1 + c)(ax_2 + by_2 + c) < 0$
this means point A and B lies on the opposite side of line



Q. **Find range of a for which (a, a^2) and origin lie on same side of line $4x+4y-3=0$**

Sol. $(4(0)+4(0)-3)(4a+4a^2-3)>0$

$\Rightarrow (4a^2+4a-3)<0$

$\Rightarrow (4a^2+6a-2a-3)<0$

$\Rightarrow 2a(2a+3)-1(2a+3)<0$

$\Rightarrow (2a-1)(2a+3)<0$

$\Rightarrow a\in\left(-\frac{3}{2},\frac{1}{2}\right)$

Q. **If point (a, a^2) lies between lines $x+y-2=0$ and $4x+4y-3=0$. Find the range of a.**

Sol. $L_1 : x+y-2=0$

$L_2 : 4x+4y-3=0$

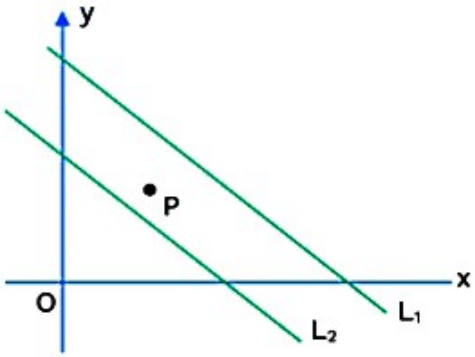
for L_1 points O and P are same side

$(0+0-2)(a+a^2-2)>0$

$(a^2+a-2)<0 \Rightarrow (a^2+2a-a-2)<0$

$a(a+2)-1(a+2)<0\ (a-1)(a+2)<0$

$\Rightarrow a\in(-2,1)$ ... (1)



for L_2 points O and P are opposite side

$(4(0)+4(0)-3)(4a+4a^2-3)<0$

$\Rightarrow (4a^2+4a-3)<0$

$\Rightarrow (4a^2+6a-2a-3)>0$

$\Rightarrow 2a(2a+3)-1(2a+3)>0$

$\Rightarrow (2a-1)(2a+3)>0$

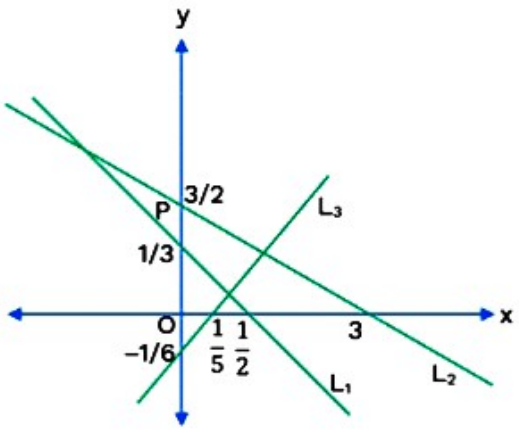
$\Rightarrow (-\infty,-3/2)\cup(1/2,\infty)$ ... (2)

Now by (1) $\cap$ (2)

$a\in\left(-2,\frac{-3}{2}\right)\cup\left(\frac{1}{2},1\right)$

Q. **Determine values of α for which point (α, α^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 3 = 0$ and $5x - 6y - 1 = 0$**

Sol. $L_1 : 2x + 3y - 1 = 0$
$L_2 : x + 2y - 3 = 0$
$L_3 : 5x - 6y - 1 = 0$



O & P lie opposite side of L_1 hence,

$(-1)(2\alpha + 3\alpha^2 - 1) < 0$

$3\alpha^2 + 2\alpha - 1 > 0 \quad 3\alpha^2 + 3\alpha - \alpha - 1 > 0$

$3\alpha(\alpha + 1) - (\alpha + 1) > 0 \Rightarrow (3\alpha - 1)(\alpha + 1) > 0$

$\alpha \in (-\infty, -1) \cup (1/3, \infty)$ ... (1)

O and P lie same side of L_2 hence

$(-3)(\alpha + 2\alpha^2 - 3) > 0 \Rightarrow 2\alpha^2 + \alpha - 3 < 0$

$2\alpha^2 + 3\alpha - 2\alpha - 3 < 0$

$\alpha(2\alpha + 3) - (2\alpha + 3) < 0 \Rightarrow (\alpha - 1)(2\alpha + 3) < 0$

$\alpha \in (-3/2, 1)$ ... (2)

O and P lies same side of L_3 hence

$(-1)(5\alpha - 6\alpha^2 - 1) > 0 \Rightarrow 6\alpha^2 - 5\alpha + 1 > 0$

$6\alpha^2 - 2\alpha - 3\alpha + 1 > 0$

$2\alpha(3\alpha - 1) - (3\alpha - 1) > 0 \Rightarrow (2\alpha - 1)(3\alpha - 1) > 0$

$\alpha \in (-\infty, 1/3) \cup (1/2, \infty)$ ... (3)

(1) $\cap$ (2) $\cap$ (3)

$\alpha \in (-3/2, -1) \cup (1/2, 1)$

Condition of concurrency :

Let the equation of the three lines be

$a_1x + b_1y + c_1 = 0$ ...(i)

$a_2x + b_2y + c_2 = 0$ ...(ii)

$a_3x + b_3y + c_3 = 0$ ...(iii)

Let the lines (i) and (ii) intersect at the point (x_1, y_1) then (x_1, y_1) will satisfy both so

$a_1x_1 + b_1y_1 + c_1 = 0$

$a_2x_1 + b_2y_1 + c_2 = 0$

point of intersection, we get

$$\frac{x_1}{b_1c_2 - b_2c_1} = \frac{y_1}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$x_1 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \text{ and } y_1 = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

Three lines are said to be concurrent if they passes through a common point.

Thus (x_1, y_1) should lie on the (iii) line for the cond. of concurrency.

$$a_3\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}\right) + b_3\left(\frac{b_1c_2 - c_2a_1}{a_1b_2 - a_2b_1}\right) + c_3 = 0$$

Which given $a_3(b_1c_2 - b_2c_1) + b_3(c_1a_2 - c_2a_1) + c_3(a_1b_2 - a_2b_1) = 0$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Note :

For any three parallel lines this condition holds but they are not concurrent.

Q. **Find k if lines $x - y = 3$, $x + y = 7$, $kx + 3y = 4$ are concurrent.**

Sol.

$$\begin{vmatrix} 1 & -1 & -3 \\ 1 & 1 & -7 \\ k & 3 & -4 \end{vmatrix} = 0$$

$1(17) + 1(7k - 4) - 3(3 - k) = 0$

$10k + 4 = 0 \Rightarrow k = -2/5$

Q. **Prove that in any triangle, altitudes are concurrent.**

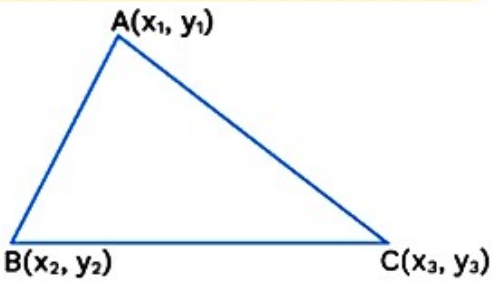
Sol.

$$m_{BC} = \frac{y_3 - y_2}{x_3 - x_2}$$

equation of altitude from A

$$(y - y_1) = -\frac{(x_3 - x_2)}{(y_3 - y_2)}(x - x_1)$$

$(y_3 - y_2)y - y_1(y_3 - y_2) = -(x_3 - x_2)x + x_1(x_3 - x_2)$



Sol.

$$\begin{vmatrix} \sin3\theta & a & b \\ \cos2\theta & c & d \\ 2 & a+2c & b+2d \end{vmatrix} = 0$$

$R_3 \to R_3 - (R_1 + 2R_2)$

$$\begin{vmatrix} \sin3\theta & a & b \\ \cos2\theta & c & d \\ 2-\sin3\theta-2\cos2\theta & 0 & 0 \end{vmatrix} = 0$$

Now

$(2 - \sin3\theta - 2\cos2\theta)(ad - bc) = 0$

$\Rightarrow 2\left(1-\cos2\theta\right) - \sin3\theta = 0$

$\Rightarrow 2\left(2\sin^2\theta\right) - 3\sin\theta + 4\sin^3\theta = 0$

$\Rightarrow \sin\theta\left\{4\sin^2\theta + 4\sin\theta - 3\right\} = 0$

$\Rightarrow \sin\theta\{4\sin^2\theta + 6\sin\theta - 2\sin\theta - 3] = 0$

$\Rightarrow \sin\theta\{2\sin\theta\left(2\sin\theta+3\right) - \left(2\sin\theta+3\right)] = 0$

$\Rightarrow \sin\theta\left(2\sin\theta-1\right)\left(2\sin\theta+3\right) = 0$

$\sin\theta = 0$	or	$\sin\theta = 1/2$	or	$\sin\theta = -3/2$
$\theta = n\pi$		$\theta = n\pi + (-1)^n \frac{\pi}{6}$		X
$(n \in I)$		$(n \in I)$		

Family of Lines

(i) If L_1 and L_2 are two concurrent lines then family of concurrent lines passing through point of intersection of L_1 and L_2 can be represented as

$L_1 + \lambda L_2 = 0$

(ii) Family of parallel lines

$ax + by + \lambda = 0$

Type – 1

Equation of line always passing through point of intersection of $l_1 = 0$ and $l_2 = 0$ is

$l_1 + \lambda l_2 = 0 \ \forall \ \lambda \in R$

Q. **Find equation of line passing through intersection of $3x-4y+6=0$ and $x+y+2=0$ and**
(i) Parallel to line $y=0$
(ii) Parallel to line $x=7$
(iii) At a distance of 5 units from origin
(iv) Situated at maximum distance from point (2,3)

Sol. Equation of required line is
$(3x-4y+6)+\lambda(x+y+2)=0$
$(3+\lambda)x+(\lambda-4)y+2(\lambda+3)=0$

(i) for this $3+\lambda=0 \Rightarrow \lambda=-3$
$-7y+0=0 \Rightarrow y=0$

(ii) for this $\lambda-4=0 \Rightarrow \lambda=4$
eq': $7x+14=0 \Rightarrow x+2=0$

(iii) $$\left|\frac{(3+\lambda)(0)+(\lambda-4)(0)+2(\lambda+3)}{\sqrt{(3+\lambda)^2+(\lambda-4)^2}}\right|=5$$

$$|2(\lambda+3)|=5\sqrt{2\lambda^2-2\lambda+25} \text{ on squaring } 4(\lambda^2+6\lambda+9)=25(2\lambda^2-2\lambda+25)$$

$$\Rightarrow 46\lambda^2-74\lambda+589=0$$

$D<0 \Rightarrow$ not possible

(iv) point of intersection of given lines is A(–2, 0)
now SL passing through A and at max.
distance from B(2, 3) will be $\perp$ to AB
hence $m_{AB}=\frac{3}{4}$

$\therefore$ slope of line is –4/3 equation : $(y-0)=-\frac{4}{3}(x+2) \Rightarrow 3y=-4x-8$

$\Rightarrow 4x+3y+8=0$

Type – 2 (Converse of Type – 1)

$l_1+\lambda l_2=0$ always passes through intersection of $l_1=0$ and $l_2=0$

Q. **Find the point through which the line $x-1+\lambda y=0$ always passes for all $\lambda \in R$.**

Sol. Given family of line is $(x-1)+\lambda(y)=0=L_1+\lambda L_2=0$

Where $L_1: x=1$
$L_2: y=0$
which always passes through point of intersection of L_1 and L_2 which is (1, 0)

Q. **Find the point through which the line $x - 2 + \lambda(y - 1) = 0$ always passes for all $\lambda \in R$.**

Sol. Given family of line is $(x-2)+\lambda(y-1)=0 = L_1 + \lambda L_2 = 0$

Where $L_1 : x = 2$

$L_2 : y = 1$

which always passes through point of intersection of L_1 and L_2 which is (2,1)

Q. **Find the point through which the line $2x - 3\lambda y = y + 7$ always passes $\forall\ \lambda \in R$.**

Sol. Given family of line is

$(2x - y - 7) - 3\lambda(y) = 0 \equiv L_1 + KL_2 = 0$

where $L_1 : 2x - y - 7 = 0$

$L_2 : y = 0$

which always passes through point of intersection of L_1 and L_2 which is (7/2,0)

Q. **If a, b, c are in AP find the point through which $ax + by + c = 0$ always passes through.**

Sol. **Method-I:**

$2b = a + c \Rightarrow c = 2b - a$

Now equation $ax + by + 2b - a = 0$

$a(x - 1) + b(y + 2) = 0$

$$\underbrace{x-1}_{L1} + \frac{b}{a}\underbrace{(y+2)}_{L_2}$$

Fixed point

(1, –2)

Method-II:

$\because\ a - 2b + c = 0$ ... (1)

$ax + by + c = 0$ ... (2)

On comparing (1) & (2)

$$\frac{x}{1} = \frac{y}{-2} = \frac{1}{1}$$

$\Rightarrow x = 1, y = -2$

fixed point (1, –2)

Q. **If a + 3b = 5c find the fixed point through which line ax + by + c=0 passes.**

Sol. $a + 3b - 5c = 0$

$ax + by + c = 0$

on comparison

$$\frac{x}{1} = \frac{y}{3} = \frac{1}{-5} \Rightarrow (1)$$

fixed point (-1/5, -3/5)

Q. **If $a^2 + 9b^2 = 6ab + 4c^2$ the $ax + by + c = 0$ passes through one or the other of two fixed point. Find the points.**

Sol. $a^2 - 6ab + 9b^2 - 4c^2 = 0$

$(a - 3b)^2 - (2c)^2 = 0 \left(a - 3b + 2c\right)\left(a - 3b - 2c\right) = 0$

Case – I : $a - 3b + 2c = 0$

$ax + by + c = 0$

$$\Rightarrow \frac{x}{1} = \frac{y}{-3} = \frac{1}{2} \text{ fixed point} \left(\frac{1}{2}, -\frac{3}{2}\right)$$

Case – II : $a - 3b - 2c = 0$

$ax + by + c = 0$

$$\Rightarrow \frac{x}{1} = \frac{y}{-3} = \frac{1}{-2} \text{ fixed point} \left(\frac{-1}{2}, \frac{3}{2}\right)$$

Q. **If algebraic sum of the perpendiculars from $A(x_1, y_1)$, $B(x_2, y_2)$, $A(x_3, y_3)$ on a variable line $ax + by + c = 0$ vanished then the variable line always passes through**

(A) G of ΔABC **(B) O of ΔABC** **(C) I of ΔABC** **(D) H of ΔABC**

Sol. **(A)**

⊥ distance from A

$$d_1 = \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \text{ (algebraic)}$$

⊥ distance from B

$$d_2 = \frac{ax_2 + by_2 + c}{\sqrt{a^2 + b^2}} \text{ (algebraic)}$$

⊥ distance from C $d_3 = \frac{ax_3 + by_3 + c}{\sqrt{a^2 + b^2}}$ (algebraic)

given $d_1 + d_2 + d_3 = 0$

$$\frac{1}{\sqrt{a^2+b^2}}\left\{a\left(x_1+x_2+x_3\right)+b\left(y_1+y_2+y_3\right)+3c\right\}=0$$

$$a\left(\frac{x_1+x_2+x_3}{3}\right)+b\left(\frac{y_1+y_2+y_3}{3}\right)+c=0$$

$\therefore$ line passes from G of ΔABC

Q. **The family of lines x(a + 2b) + y(a + 3b) = a + b always passes through a fixed point. Find the point.**

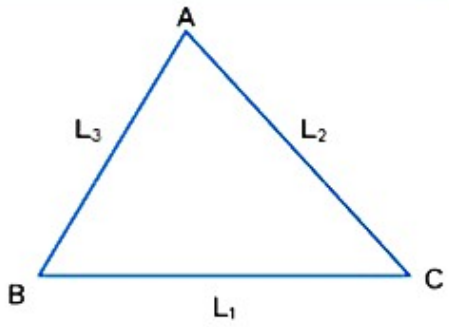
Sol. $a(x + y - 1) + b(2x + 3y - 1) = 0$

$$\underbrace{x+y-1}_{L1}+\frac{b}{a}\underbrace{(2x+3y-1)}_{L_2}=0$$

which always passes through point of int. of L_1 and L_2 which is (2, –1)

Q. **The equations to the sides of a triangle are x+2y=0,4x + 3y = 5 and 3x+y=0. Find the co-ordinates of the orthocentre of the triangle without finding vertices of triangle**

Sol. $L_1 : x + 2y = 0$
$L_2 : 4x + 3y = 5$
$L_3 : 3x + y = 0$



Altitude from A is
$L_2 + \lambda L_3 = 0$
$(4x + 3y - 5) + \lambda(3x + y) = 0$
$(4 + 3\lambda)x + (\lambda + 3)y - 5 = 0$

which is $\perp$ to L_1 hence $-\left(\frac{4+3\lambda}{3+\lambda}\right)\left(\frac{-1}{2}\right) = -1$

$4 + 3\lambda = -6 - 2\lambda \Rightarrow \lambda = -2$
$-2x + y - 5 = 0 \Rightarrow 2x - y + 5 = 0$ ... (1)

altitude from B is
$L_1 + \mu L_3 = 0$
$(x + 2y) + \mu(3x + y) = 0$
$(3\mu + 1)x + (\mu + 2)y = 0$
Which is $\perp$ to L_2 hence

$$-\left(\frac{3\mu+1}{\mu+2}\right)\left(-\frac{4}{3}\right)=-1$$

$12\mu + 4 = -3\mu - 6 \Rightarrow \mu = -2/3$

equation $-x + 4y/3 = 0 \Rightarrow 3x = 4y$ ... (2)

On solving (1) and (2)

(-4 , -3) orthocentre

Type – 3

Equation of diagonal of parallelogram

If lines of parallelogram are considered as

$\mu_1 = a_1x + b_1y + c_1 = 0$

$\mu_2 = a_2x + b_2y + d_1 = 0$

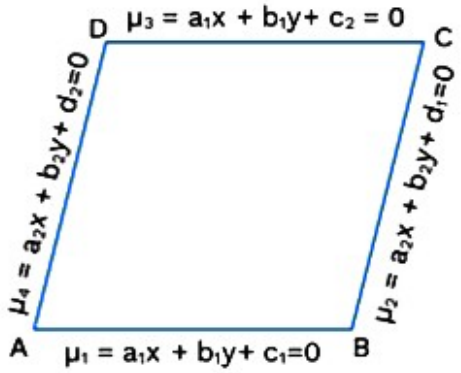
$\mu_3 = a_1x + b_1y + c_2 = 0$

$\mu_4 = a_2x + b_2y + d_2 = 0$

then equations of diagonals will be

$AC : \mu_1\mu_2 - \mu_3\mu_4 = 0$

$BD : \mu_1\mu_4 - \mu_2\mu_3 = 0$



Q. **Find the equations of the diagonals of the parallelogram formed by the lines**
$2x - y + 7 = 0$, $2x - y - 5 = 0$
$3x + 2y - 5 = 0$, $3x + 2y + 7 = 0$

Sol. Equation of diagonal -1 is

$$\underbrace{(2x - y}_{\alpha} + 7)\underbrace{(3x + 2y}_{\beta} - 5) - \underbrace{(2x - y}_{\alpha} - 5)\underbrace{(3x + 2y}_{\beta} + 7) = 0$$

$(\alpha\beta - 5\alpha + 7\beta - 35) - (\alpha\beta + 7\alpha - 5\beta - 35) = 0$

$12\beta = 12\alpha$

$3x + 2y = 2x - y \Rightarrow x + 3y = 0$

Equation of diagonal -2 is

$$\underbrace{(2x - y}_{\alpha} + 7)\underbrace{(3x + 2y}_{\beta} + 7) - \underbrace{(2x - y}_{\alpha} - 5)\underbrace{(3x + 2y}_{\beta} - 5) = 0$$

$(\alpha\beta + 7\alpha + 7\beta + 49) - (\alpha\beta - 5\alpha - 5\beta + 25) = 0$

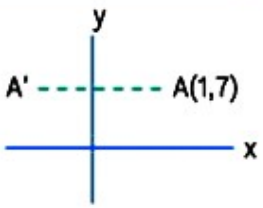
$12\alpha + 12\beta + 24 = 0 \Rightarrow \alpha + \beta + 2 = 0$

$(2x - y) + (3x + 2y) + 2 = 0 \Rightarrow 5x + y + 2 = 0$

Optics Problems

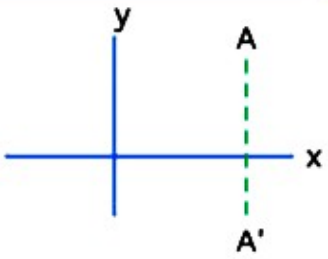
Q. **Find reflection of point A(1, 7) about y – axis.**

Sol. Reflection is A'(–1,7)

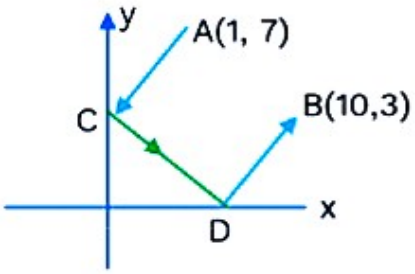


Q. **Find reflection of point (10, 3) about x – axis.**

Sol. Reflection is A'(10, –3)



Q. **If A(1, 7), B(10, 3). Find co-ordinate of point C and D.**

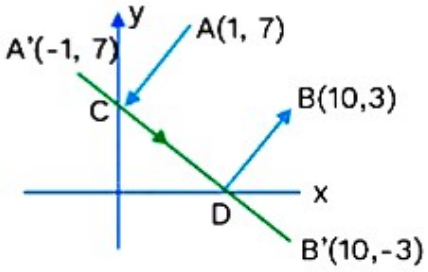


Sol. Let reflection of A in y – axis is A' and reflection of B in x – axis is B'
Hence A'(–1, 7) & B'(10, –3)
Now A', C, D, B' are collinear
hence equation

$$(y+3)=\frac{10}{-11}(x-10)$$

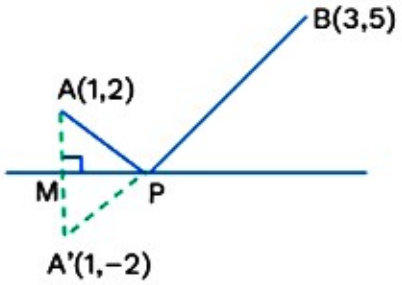
$\Rightarrow$ -11y – 33 = 10x - 100
10x + 11y = 67
Now

$$C\left(0,\frac{67}{11}\right) \text{ \& } D\left(\frac{67}{10},0\right)$$



Q. **A(1,2) and B(3,5). Point P lies on x -axis. Find P such that AP + PB is minimum.**

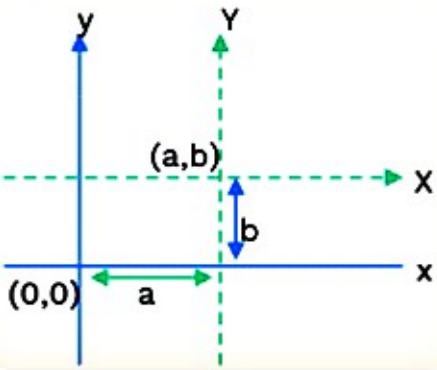
Sol. Let reflection of A in x-axis A'(1, –2)
Now $\Delta AMP \cong \Delta A'MP$
$\therefore$ AP = A'P
Now AP + PB = A'P + PB
also A'P +PB will be minimum if A',P,B will be collinear
equation of A'B
$(y + 2) = 7/2(x - 1)$
$\Rightarrow 2y + 4 = 7x - 7 \Rightarrow 7x - 2y = 11$
hence P(11/7, 0)



Shifting of origin

If origin is shifted to (a, b), then

$X = x - a$
$Y = y - b$
where (X, Y) are the coordinates with respect to new coordinate axis.



Q. **Find the new co-ordinate of point (3, –4) if origin is shifted to (1, 2)**

Sol. (x, y) ≡ (3, –4) and (a, b) ≡ (1, 2)
$X = x - a = 3 - 1 = 2$
$Y = y - b = -4 - 2 = -6$
hence new co-ordinate is (2, –6)

Q. **Find transformed equation of the straight line 2x – 3y + 5 = 0 if origin is shifted to (3,-1).**

Sol. $X = x - 3, Y = y + 1$
new equation of line is $2(X + 3) - 3(Y - 1) + 5 = 0$
$\Rightarrow 2X - 3Y + 14 = 0$

Q. **Find the point to which the origin should be shifted so that the equation $x^2 + y^2 - 5x + 2y - 5 = 0$ has no one degree terms.**

Sol. Let that point be (a, b) hence.

$X = x - a, Y = y - b$

now equation becomes

$(X + a)^2+(Y + b)^2-5(X + a) + 2(Y + b) - 5 = 0$

$(X^2+2aX+a^2) + (Y^2 + 2bY + b^2) -5X - 5a + 2Y + 2b - 5 = 0$

$X^2+Y^2+(2a - 5)X + (2b + 2)Y + (a^2 - 5a + b^2 + 2b - 5) = 0$

which should not have one degree terms

hence

$2a - 5 = 0$ & $2b + 2 = 0$

$a = 5/2$ & $b = -1 \Rightarrow$ Point $(5/2, -1)$

Q. **Find the point to which the origin should be shifted so that the equation $y^2 - 6y - 4x + 13 = 0$ is transformed to $y^2 = Ax$**

Sol. Let that point be (a, b) hence

$X = x - a, Y = y - b$

now equation becomes

$(Y + b)^2 - 6(Y + b) - 4(X + a) + 13 = 0$

$Y^2 + 2bY + b^2 - 6Y - 6b - 4X - 4a + 13 = 0$

$Y^2 + (2b - 6)Y + (b^2 - 6b - 4a + 13) - 4X = 0$

for required equation

$2b - 6 = 0$ & $b^2 - 6b - 4a + 13 = 0$

$b = 3$ & $9 - 18 - 4a + 13 = 0$

$4 = 4a \Rightarrow a = 1$

point is (1, 3)

Q. **Find area of triangle formed with vertices (2, 0), (0, 0),(1, 4) if origin is shifted to (2010, 2012)**

Sol. After shifting the origin length of sides of triangle remain same hence area will also remain same. hence area

$$\Delta = \frac{1}{2}\left\|\begin{vmatrix}2 & 0\\ 0 & 0\end{vmatrix} + \begin{vmatrix}0 & 1\\ 0 & 4\end{vmatrix} + \begin{vmatrix}1 & 2\\ 4 & 0\end{vmatrix}\right\|$$

$\frac{1}{2}|0 + 0 - 8| = 4$ sq. unit

Note :

Slope of line remains same after changing the origin.

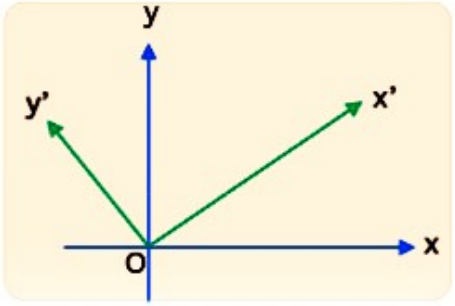
Q. **If the axes are shifted to (1,1) then what do the following becomes**
(i) $x^2 + xy - 3y^2 - y + 2 = 0$
(ii) $xy - x - y + 1 = 0$
(iii) $x^2 - y^2 - 2x + 2y = 0$

Sol. **(i)** $X = x - 1, Y = y - 1$
$x = X + 1, y = Y + 1$
new equation : $(X + 1)^2 + (X + 1)(Y + 1) - 3(Y + 1)^2 - (Y + 1) + 2 = 0$
$\Rightarrow (X^2 + 2X + 1) + (XY + X + Y + 1) - 3(Y^2 + 2Y + 1) - Y - 1 + 2 = 0$
$\Rightarrow X^2 + XY - 3Y^2 + 3X - 6Y = 0$

(ii) new equation is
$(X + 1)(Y + 1) - (X + 1) - (Y + 1) + 1 = 0$
$XY + X + Y + 1 - X - 1 - Y - 1 + 1 = 0$
$XY = 0$

(iii) $(X^2 - 2X + 1) - (Y^2 - 2Y + 1) = -1$
$(X - 1)^2 - (Y - 1)^2 = 0$
$X^2 - Y^2 = 0$

Rotation of Co-ordinate system



$OA = OC - AC$

$x = X\cos\theta - Y\sin\theta$ ($\because$ AC = NB)

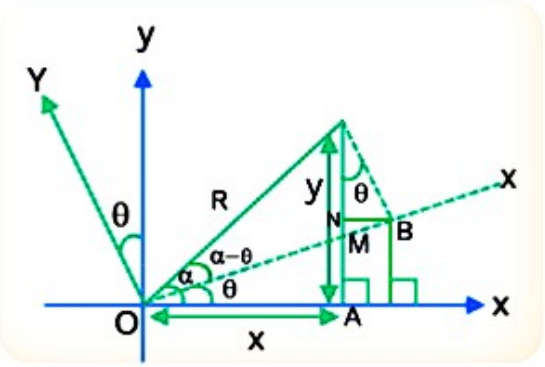
where (X, Y) are the coordinates with respect to

new coordinate system.

$PA = AM + PM$

$y = X\sin\theta + Y\cos\theta$

	x	y
X	$\cos\theta$	$\sin\theta$
Y	$-\sin\theta$	$\cos\theta$



Q. **If the axes are rotated through an angle of 30° in the anticlockwise direction about the origin. The co-ordinates of point are $\left(4,-2\sqrt{3}\right)$ in the new system.**

Find its old co-ordinates.

Sol. Given $\theta = 30°$

in new system $X = 4, Y = -2\sqrt{3}$

in old system $x = X\cos\theta - Y\sin\theta$

$= 4\cos 30^0 + 2\sqrt{3}\sin 30^0$

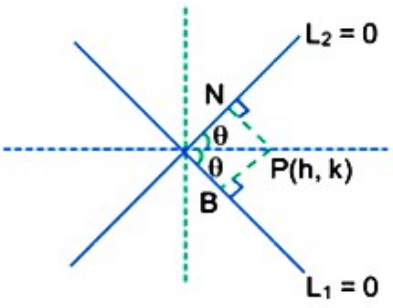
$= 4\left(\frac{\sqrt{3}}{2}\right) + 2\sqrt{3}\left(\frac{1}{2}\right) = 3\sqrt{3}$

$y = X\sin\theta + Y\cos\theta$

$= 4\sin 30° - 2\sqrt{3}\cos 30° = 4\left(\frac{1}{2}\right) - 2\sqrt{3}\left(\frac{\sqrt{3}}{2}\right) = -1$, old co-ordinates are $\left(3\sqrt{3}, -1\right)$

Angle Bisector

The bisectors of the angle between the lines $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$ is the locus of the point which moves such that the lengths of perpendiculars drawn from it to the two given lines are equal



By definition PN = PB

$$\left|\frac{a_1h + b_1k + c_1}{\sqrt{a_1^2 + b_1^2}}\right| = \left|\frac{a_2h + b_2k + c_2}{\sqrt{a_2^2 + b_2^2}}\right|$$

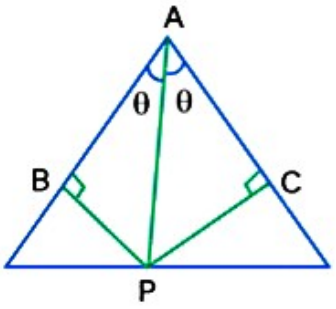
$$\frac{a_1h + b_1k + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2h + b_2k + c_2}{\sqrt{a_2^2 + b^2}}$$

hence the locus

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b^2}}$$

PB = PC

$\Delta ABP \cong \Delta ACP$



To discriminate between the bisector of the angle containing the origin and that of the angle not containing the origin:

Proof:

To isolate between the bisector of the pair containing and not containing the origin.

Let $L_1 \equiv a_1x + b_1y + c_1 = 0$

$L_2 \equiv a_2x + b_2y + c_2 = 0$

P and O are the same side of $L_2 = 0$

Since c_1 is positive so $a_1x + b_1y + c_1$ is positive.

c_2 is positive so $a_2x + b_2y + c_2$ is positive.

So, $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

bisector of angle that contains the origin

Now (0, 0) and (h, k) lies on the same side of L_1 but opposite side of L_2 so $a_1h + b_1k + c_1 > 0$ and $a_2h + b_2k + c_2 < 0$

Hence $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the required bisector (does not contains origin)

To discriminate between the bisector of angle containing origin and that of the angle not containing origin.

(i) Rewrite lines with same sign of absolute term

(ii) Now positive sign will given you origin containing angle bisector.

To discriminate between acute angle bisector and obtuse angle bisector:

Write $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ such that constant terms is positive.

If $a_1a_2 + b_1b_2 < 0$ then the angle between the lines contains origins is acute and the equation of bisector of this acute angle is

$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$, $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of the other bisector.

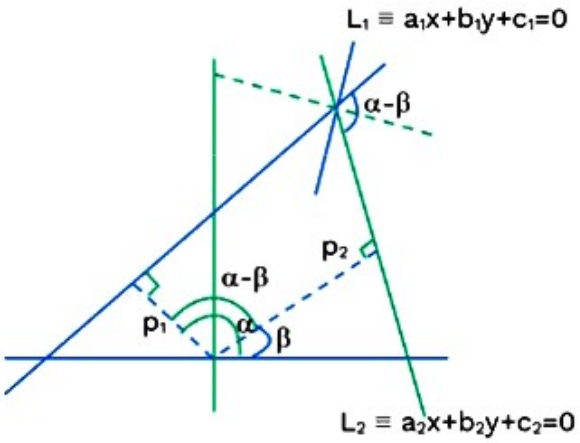
If $a_1a_2 + b_1b_2 > 0$ then the angle between the lines that contains the origin is obtuse and the equation of the bisector of this obtuse angle is $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

Therefore:

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = -\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

The equation of the other bisector

Proof: $a_1x + b_1y + c_1 = 0$ (c_1 is positive)



$a_1x + b_1y = -c_1$

$-a_1x - b_1y = c_1$

$$\frac{-a_1}{\sqrt{a_1^2+b_1^2}} - \frac{b_1}{\sqrt{a_1^2+b_1^2}} = \frac{c_1}{\sqrt{a_1^2+b_1^2}} \quad \ldots (1)$$

Let the normal form of (1) is

$x\cos\alpha + y\sin\alpha = p_1$

Comparing

$$\cos\alpha = \frac{-a_1}{\sqrt{a_1^2+b_1^2}} \text{ and } \sin\alpha = \frac{-b_1}{\sqrt{a_1^2+b_1^2}}$$

Similarly,

$$\cos\beta = \frac{-a_2}{\sqrt{a_2^2+b_2^2}} \text{ and } \sin\beta = \frac{-b_2}{\sqrt{a_2^2+b^2}}$$

$$\cos\alpha\cos\beta + \sin\alpha\sin\beta = \frac{a_1a_2}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}} + \frac{b_1b_2}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}$$

$$\cos(\alpha-\beta) = \frac{a_1a_2+b_1b_2}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}$$

If $a_1a_2 + b_1b_2 < 0 \Rightarrow \cos(\alpha - \beta) < 0 \Rightarrow (\alpha - \beta)$ is obtuse

This means origin lies in the acute region $\Rightarrow$ positive sign bisector will give the bisector of the pair which contains the origin as well as acute angle bisector and positive sign will give the other bisector.

If $a_1a_2 + b_1b_2 > 0 \Rightarrow \cos(\alpha - \beta) > 0 \Rightarrow (\alpha - \beta)$ is acute. This means origin lies in the obtuse region. Hence in this case positive sign bisector will be the obtuse angle bisector that contains the origin.

To discriminate between the acute and obtuse angle bisector (Method-1)

If lines are $L_1 : a_1x + b_1y + c_1 = 0$ and (after making c_1, c_2 sign same)

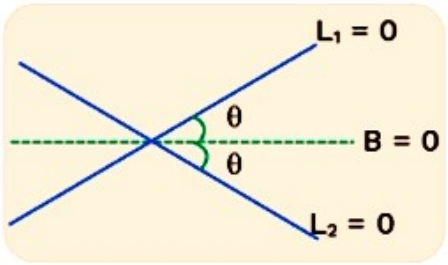
$L_2 : a_2x + b_2y + c_2 = 0$

and

(i) $a_1a_2 + b_1b_2 > 0 \Rightarrow$ (+) sign gives obtuse A.B.
(–) sign given acute A.B

(ii) $a_1a_2 + b_1b_2 < 0 \Rightarrow$ (+) sign gives acute A.B.
(–) sign gives obtuse A.B.

To discriminate between the acute & obtuse angle bisector (method-2)



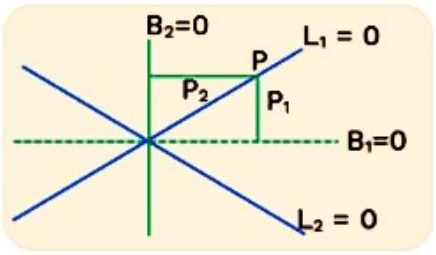
If θ is acute angle between bisector B = 0 and line L_1 = 0

Now if

$|\tan\theta| < 1 \Rightarrow B = 0$ is acute angle bisector

$|\tan\theta| > 1 \Rightarrow B = 0$ is obtuse angle bisector

To discriminate between the acute & obtuse angle bisector (Method-3)



If P_1 and P_2 are length of perpendiculars from point P on $L_1 = 0$ on bisectors $B_1 = 0$ and $B_2 = 0$ respectively

Now if

$P_1 > P_2 \Rightarrow B_1 = 0$ is obtuse angle bisector

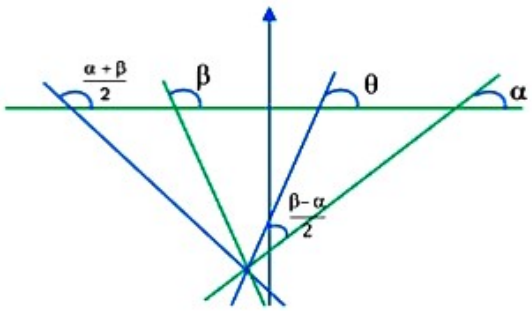
$P_1 < P_2 \Rightarrow B_1 = 0$ is acute angle bisector

Bisectors between lines expressed in parameter

form $\frac{x - x_1}{\cos\alpha} = \frac{y - y_1}{\sin\alpha}$ and $\frac{x - x_1}{\cos\beta} = \frac{y - y_1}{\sin\beta}$

$$\frac{x - x_1}{\cos\frac{\alpha+\beta}{2}} = \frac{y - y_1}{\sin\frac{\alpha+\beta}{2}}$$

and $$\frac{x - x_1}{-\sin\frac{\alpha+\beta}{2}} = \frac{y - y_1}{\cos\frac{\alpha+\beta}{2}}$$



hence $\theta = \frac{\beta-\alpha}{2} + \alpha = \frac{\alpha+\beta}{2}$

Q. **Find the equation of bisectors of the lines**
$4x + 3y - 7 = 0$
$24x + 7y - 31 = 0$
and also find the bisector containing origin?

Sol. $4x + 3y - 7 = 0 \quad \Rightarrow -4x - 3y + 7 = 0$
$24x + 7y - 31 = 0 \Rightarrow -24x - 7y + 31 = 0$

Equation of bisectors:

$$\frac{-4x - 3y + 7}{5} = \pm\frac{-24x - 7y + 31}{25}$$

$$-\frac{4x - 3y + 7}{1} = \pm\frac{-24x - 7y + 31}{25}$$

$-20x - 15y + 35 = -24x - 7y + 31$

and $-20x - 15y + 35 = +24x + 7y - 31$

$4x - 8y + 4 = 0$ and $44x + 22y - 66 = 0$

$x - 2y + 1 = 0$ (origin containing)

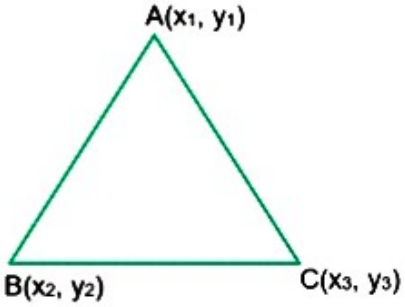
and $2x + y - 3 = 0$

Now $(-4)(-24) + (-3)(-7) > 0$

$x - 2y + 1 = 0 \rightarrow$ origin containing obtuse angle bisector

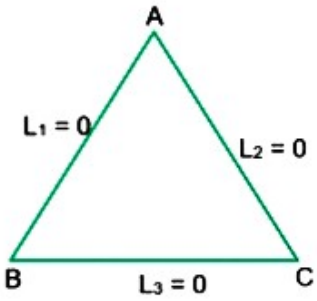
Bisector in case of triangle

(i) When vertices of triangle are known find incentre then all three angle bisectors.



(ii) When sides are known compute tanA, tanB, tanC by arranging the lines in descending order of their slopes.

Now compute acute obtuse angle bisectors as the case.



Q. **Find equations of angle bisectors for lines $3x + 4y + 1 = 0$, $12x + 5y + 3 = 0$**

Sol. Equation is $\frac{3x + 4y + 1}{5} = \pm \frac{12x + 5y + 3}{13}$

$\Rightarrow 39x + 52y + 13 = \pm (60x + 25y + 15)$

(+) sign

$39x + 52y + 13 = 60x + 25y + 15$

$\Rightarrow 21x - 27y + 2 = 0$

(−) sign

$39x + 52y + 13 = -(60x + 25y + 15)$

$\Rightarrow 99x + 77y + 28 = 0$

Q. **Find the bisectors between the line $4x + 3y - 7 = 0$ and $24x + 7y - 31 = 0$. Identify acute / obtuse and origin containing / not containing.**

Sol. Equation of bisectors

$$\frac{4x + 3y - 7}{5} = \pm \frac{(24x + 7y - 31)}{25}$$

$20x + 15y - 35 = \pm (24x + 7y - 31)$

(+) sign

$B_1 : 4x - 8y + 4 = 0$

$B_1 : x - 2y + 1 = 0$

(−) sign

$B_2 : 44x + 22y = 66$

$B_2 : 2x + y = 3$

now for acute and obtuse angle bisector

$a_1a_2 + b_1 b_2 = 4(24) + 3(7) > 0$

$\Rightarrow B_1 = 0$ obtuse angle bisector

$\Rightarrow B_2 = 0$ acute angle bisector

Now for origin containing / Not containing.

$C_1 C_2 = (-7)(-31) > 0$

$\Rightarrow$ (+) sign $B_1 = 0$ is origin containing

(−) sign $B_2 = 0$ is non origin containing.

Q. **The vertices of a ΔABC are $A(-1, 11)$, $B(-9, -8)$ and $C(15, -2)$. Find the equation of the bisector of the angle at vertex A.**

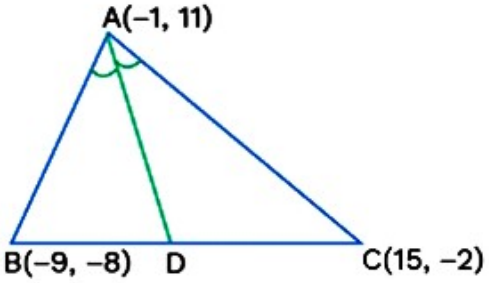
Sol. $AB = c = \sqrt{8^2 + 19^2} = 5\sqrt{17}$

$AC = b = \sqrt{16^2 + 13^2} = 5\sqrt{17}$

Point D will divide BC in ratio

$AB : AC = 1 : 1$

$\therefore D\left(\frac{-9+15}{2}, \frac{-8-2}{2}\right) \equiv D(3, -5)$



equation of bisector AD

$(y + 5) = \left(\frac{11+5}{-1-3}\right)(x - 3)$

$(y + 5) = -4x + 12 \Rightarrow 4x + y = 7$

Q. **Find bisectors between the lines $x + \sqrt{3}y = 6 + 2\sqrt{3}$ and $x - \sqrt{3}y = 6 - 2\sqrt{3}$**

Sol. $\frac{x + \sqrt{3}y - 6 - 2\sqrt{3}}{2} = \pm \frac{\left(x - \sqrt{3}y - 6 + 2\sqrt{3}\right)}{2}$

(+) sign $B_1 : 2\sqrt{3}y = 4\sqrt{3}$

$B_1 : y = 2$

(–) sign $B_2 : 2x = 12$

$B_2 : x = 6$

Note :

If $m_1 + m_2 = 0$ lines equally inclined with the axes.

Pair of straight line

Pair of line through origin.

$ax^2 + 2hxy + by^2 = 0$ (2° equation)

(i) $h^2 > ab \Rightarrow$ lines are real & distinct

(ii) $h^2 = ab \Rightarrow$ lines are coincidental

(iii) $h^2 < ab \Rightarrow$ lines are imaginary with real point of intersection

Proof:

Given equation is $b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$

$D = 4h^2 - 4ba = 4(h^2 - ab)$

$D > 0 \Rightarrow$ lines are real & distinct

$D = 0 \Rightarrow$ lines are coincidental

$D < 0 \Rightarrow$ lines are imaginary

Note :

(i) A homogeneous equation of degree 'n' represents 'n straight lines passing through origin.

(ii) If $y = m_1x$, $y = m_2x$ be two equation represented by $ax^2 + 2hxy + by^2 = 0$ then

$$m_1 + m_2 = -\frac{2h}{b}, m_1m_2 = \frac{a}{b}$$

Angle between two lines $ax^2 + 2hxy + by^2 = 0$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right|$$

Lines being perpendicular
coefficient of x^2 + coefficient of $y^2 = 0$
i.e. $a + b = 0$
Lines are parallel or coincident
$h^2 = ab$
Lines are equally inclined to x-axis then coordinate axes are bisectors.
coefficient of $xy = 0$

Q. Find angle between lines given by $x^2 + 4xy + 4y^2 = 0$

Sol. $a = 1, h = 2, b = 4$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \left|\frac{2\sqrt{4-4}}{5}\right| = 0$$

$\Rightarrow \theta = 0$

Q. Find angle between lines given by $x^2 + 4xy + y^2 = 0$

Sol. $a = 1, h = 2, b = 1$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \left|\frac{2\sqrt{4-1}}{2}\right| = \sqrt{3}$$

$\Rightarrow \theta = \pi/3$

Q. Find angle between lines given by $y^2 - 3x^2 = 0$

Sol. $a = -3, h = = 0, b = 1$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \left|\frac{2\sqrt{0+3}}{-2}\right| = \sqrt{3}$$

$\Rightarrow \pi/3$

Q. **Find angle between lines given by $xy = 0$**

Sol. The lines are $x = 0$ (y-axis)
and $y = 0$ (x – axis)
$\Rightarrow$ angle between them $= \pi/2$

Q. **Find angle between lines given by $3x^2 + 10xy - 3y^2 = 0$**

Sol. $a = 3, h = 5, b = -3$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \left|\frac{2\sqrt{25+9}}{0}\right|$$

$\Rightarrow \tan\theta$ is not defined hence $\theta = \pi/2$

General Equation of 2°

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
Condition that 2° equation represents pair of lines
$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

or $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

Q. **Find whether $2x^2 - xy - y^2 - x + 4y - 3$, can be factorized in two linears. If yes find the factors. Also find the angle between the lines.**

Sol. $a = 2, h = -1/2, b = -1, g = -1/2, f = 2, c = -3$
$abc + 2fgh - af^2 - bg^2 - ch^2$
$= 2(-1)(-3) + 2(2)(-1/2)(-1/2) - 2(2)^2 - (-1)(-1/2)^2 - (-3)(-1/2)^2$
$\Rightarrow 6 + 1 - 8 + 1/4 + 3/4 = 0$
$\Rightarrow$ Given equation can be factorized
Now to find factors
$2x^2 - xy - y^2 = 2x^2 - 2xy + xy - y^2$
$= 2x(x - y) + y(x - y)$
$= (2x + y)(x - y)$
let factors are
$(2x + y + c_1)(x - y + c_2)$
coeff. of $x \Rightarrow c_1 + 2c_2 = -1$
coeff. of $y \Rightarrow c_2 - c_1 = 4 \Rightarrow c_1 = -3, c_2 = 1$
lines are $2x + y - 3 = 0$ & $x - y + 1 = 0$

acute angle b/w lines

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \left|\frac{2\sqrt{1/4 + 2}}{1}\right|$$

$\tan\theta = 3 \Rightarrow \theta = \tan^{-1}(3)$

Q. **$12x^2 + 7xy - 10y^2 + 13x + 45y - 35 = 0$. Factorize this in two liners.**

Sol. $12x^2 + 7xy - 10y^2 = 12x^2 + 15xy - 8xy - 10y^2$

$= 3x(4x + 5y) - 2y(4x + 5y)$

$= (3x - 2y)(4x + 5y)$

let factors are

$(3x - 2y + c_1)(4x + 5y + c_2)$

coefficient of $x \Rightarrow 4c_1 + 3c_2 = 13$ ... (1)

coefficient of $y \Rightarrow 5c_1 - 2c_2 = 45$ ... (2)

$(1) \times 2 + (2) \times 3$

$23c_1 = 161 \Rightarrow c_1 = 7$, $c_2 = -5$

lines are $3x - 2y + 7 = 0$ & $4x + 5y - 5 = 0$

Q. **Find condition for which $ax^3 + bx^2y + cxy^2 + dy^3 = 0$. Represent three lines two of which are at right angles.**

Sol. On dividing the given equation of x^3

$\Rightarrow a + b(y/x) + c(y/x)^2 + d(y/x)^3 = 0$

Let $y/x = m$ (slope)

$dm^3 + cm^2 + bm + a = 0$ ... (1)

let roots are m_1, m_2 , m_3 and given that $m_1m_2 = -1$

product of roots $m_1m_2m_3 = -a/d$

$-m_3 = -a/d \Rightarrow m_3 = a/d$

in equation (1)

$$d\left(\frac{a}{d}\right)^3 + c\left(\frac{a}{d}\right)^2 + b\left(\frac{a}{d}\right) + a = 0$$

$$\Rightarrow \frac{a^3}{d^2} + \frac{ca^2}{d^2} + \frac{ab}{d} + a = 0$$

$\Rightarrow a^3 + ca^2 + abd + ad^2 = 0$

$\Rightarrow a^2 + ac + bd + d^2 = 0$

Q. **Prove that $3x^2 - 8xy - 3y^2 = 0$ and $x + 2y = 3$ enclose a right isosceles Δ also find area of Δ.**

Sol. $3x^2 - 8xy - 3y^2 = 3x^2 - 9xy + xy - 3y^2$

$= 3x(x - 3y) + y(x - 3y)$

$= (3x + y)(x - 3y)$

Lines $L_1 : 3x + y = 0$; $m_1 = -3$

$L_2 : x - 3y = 0$; $m_2 = 1/3$

$L_3 : x + 2y - 3 = 0$; $m_3 = -\frac{1}{2}$

lines are L_1 & L_2 are $\perp \Rightarrow \Delta$ is Right angled

Now angle b/w L_1 & L_3

$$\theta = \tan^{-1}\left|\frac{m_1 - m_3}{1 + m_1 m_3}\right| = \tan^{-1}\left|\frac{-3 + 1/2}{1 + 3/2}\right|$$

$\theta = \tan^{-1}(1) = \pi/4$

hence angles of Δ are $\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{4}$

$$\text{area } p^2 = \left|\frac{0 + 2(0) - 3}{\sqrt{1+4}}\right|^2 = \frac{9}{5} \text{ sq. units}$$

Q. **Prove that the lines $x^2 - 4xy + y^2 = 0$ and $x + y = 1$ enclose an equilateral triangle. Find also its area.**

Sol. $x^2 - 4xy + y^2 = 0 \Rightarrow \left(\frac{y}{x}\right)^2 - 4\left(\frac{y}{x}\right) + 1 = 0$

$$\frac{y}{x} = \frac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}$$

hence $L_1 : y = \left(2 + \sqrt{3}\right)x$: $m_1 = 2 + \sqrt{3}$

$L_2 : y = \left(2 - \sqrt{3}\right)x$; $m_2 = 2 - \sqrt{3}$

$L_3 : x + y = 1$; $m_3 = -1$

$(m_1 > m_2 > m_3)$

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{2\sqrt{3}}{1+1} = \sqrt{3} \Rightarrow A = \frac{\pi}{3}$$

$$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{3 - \sqrt{3}}{1 - 2 + \sqrt{3}} = \frac{\sqrt{3}\left(\sqrt{3} - 1\right)}{\left(\sqrt{3} - 1\right)}$$

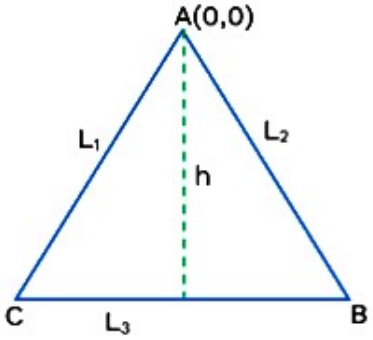
$\Rightarrow B = \pi/3$

hence Δ formed is equilateral.

Area $= \dfrac{h^2}{\sqrt{3}}$

where $h = \left|\dfrac{0+0-1}{\sqrt{1^2+1^2}}\right| = \dfrac{1}{\sqrt{2}}$

$\Rightarrow$ Area $= \dfrac{1}{2\sqrt{3}}$ sq. unit



Q. **The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point (2, 2) is :**

(A) 2 **(B)** $2\sqrt{2}$ **(C)** $4\sqrt{2}$ **(D)** $\sqrt{2}$

Sol. $x^2 + 2xy - 3y^2 = x^2 + 3xy - xy - 3y^2$

$= x(x + 3y) - y(x + 3y)$

$= (x - y)(x + 3y)$

lines are $L_1 : x - y = 0$ & $L_2 : x + 3y = 0$

$\because$ (2, 2) lies on L_1

$\therefore$ equation of Normal to L_1 at (2,2) is $x + y = 4$

$\therefore$ distance of it from origin

$= \left|\dfrac{0+0-4}{\sqrt{2}}\right| = 2\sqrt{2}$ unit.

Q. **Find centroid of Δ, the equation of whose sides are $12x^2 - 20xy + 7y^2 = 0$ and $2x - 3y + 4 = 0$**

Sol. $12x^2 - 20xy + 7y^2 = 12x^2 - 14xy - 6xy + 7y^2$

$= 2x(6x - 7y) - y(6x - 7y)$

$= (2x - y)(6x - 7y)$

Sides are $L_1 : y = 2x$

$L_2 : 7y = 6x$

$L_3 = 2x - 3y + 4 = 0$

point of intersection of sides

L_1 & L_2 is (0,0), L_2 & L_3 is (7,6)

L_1 & L_3 is (1, 2)

centroid $\left(\dfrac{8}{3}, \dfrac{8}{3}\right)$

Q. **Find distance between parallel lines**
(i) $4x^2 + 4xy + y^2 - 6x - 3y - 4 = 0$

Sol. $4x^2 + 4xy + y^2 = (2x + y)^2$
Lines are $L_1 : 2x + y + c_1 = 0$
$L_2 : 2x + y + c_2 = 0$
comparison of coeff. of $x \Rightarrow 2c_1 + 2c_2 = -6 \Rightarrow c_1 + c_2 = -3$
constant term $\Rightarrow c_1 c_2 = -4$
on solving we get $c_1 = -4$, $c_2 = 1$ (or vice-versa)
distance between L_1 and $L_2 = \left|\frac{c_1 - c_2}{\sqrt{4+1}}\right| = \frac{5}{\sqrt{5}} = \sqrt{5}$ unit

(ii) $x^2 + 2\sqrt{3}xy + 3y^2 - 3x - 3\sqrt{3}y - 4 = 0$

Sol. $x^2 + 2\sqrt{3}\,xy + 3y^2 = \left(x + \sqrt{3}\,y\right)^2$

Lines are $L_1 : x + \sqrt{3}\,y + c_1 = 0$

$L_2 : x + \sqrt{3}\,y + c_2 = 0$

On comparison of
coefficient of $x \Rightarrow c_1 + c_2 = -3$
Constant term $\Rightarrow c_1 c_2 = -4$
on solving we get $c_1 = -4$, $c_2 = 1$
(or vice-versa)
distance between L_1 and $L_2 = \left|\frac{c_1 - c_2}{\sqrt{1+3}}\right| = \frac{5}{2}$ units.

Q. **The equation $ax^2 + 6xy - 5y^2 - 4x + 6y + c = 0$ represents two perpendicular straight lines, find 'a' and 'c'**

Sol. For $\perp$ lines $a + b = 0$
$a + (-5) = 0 \Rightarrow a = 5$
now equation represents a pair of SL and
hence $\Delta = 0$
$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$
$5(-5)c + 2(3)(-2)(3) - 5(3)^2 - (-5)(-2)^2 - c(3)^2 = 0$
$-25c - 36 - 45 + 20 - 9c = 0$
$34c = -61 \Rightarrow c = -61/34$

Equation of angle bisector of

$ax^2 + 2hxy + by^2 = 0$

Proof:

Method-I

$y - m_1x = 0$ $\qquad m_1 + m_2 = -2h/b$

$y - m_2x = 0$ $\qquad m_1m_2 = a/b$

Equation of angle bisector

$$\frac{y - m_1x}{\sqrt{1+m_1^2}} = \pm\frac{y - m_2x}{\sqrt{1+m_2^2}}$$

Combined equation of angle bisectors

$$\left(\frac{y - m_1x}{\sqrt{1+m_1^2}} - \frac{y - m_2x}{\sqrt{1+m_2^2}}\right)\left(\frac{y - m_1x}{\sqrt{1+m_1^2}} + \frac{y - m_2n}{\sqrt{1+m_2^2}}\right) = 0$$

$$\frac{(y - m_1x)^2}{(1+m_1^2)} - \frac{(y - m_2x)^2}{(1+m_2^2)^2} = 0$$

$$(1+m_2^2)(y^2 - 2m_1xy + m_1^2x^2) - (1+m_1^2)(y^2 - 2m_2xy + m_2^2x_2^2) = 0$$

$$(m_1^2 - m_2^2)(x^2 - y^2) + 2(m_1m_2 - 1)(m_1m_2)xy = 0$$

$(m_1 + m_2)(x^2 - y^2) + 2(m_1m_2 - 1)xy = 0$

$$\left(-\frac{2h}{b}\right)(x^2 - y^2) + 2\left(\frac{a}{b} - 1\right)xy = 0$$

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

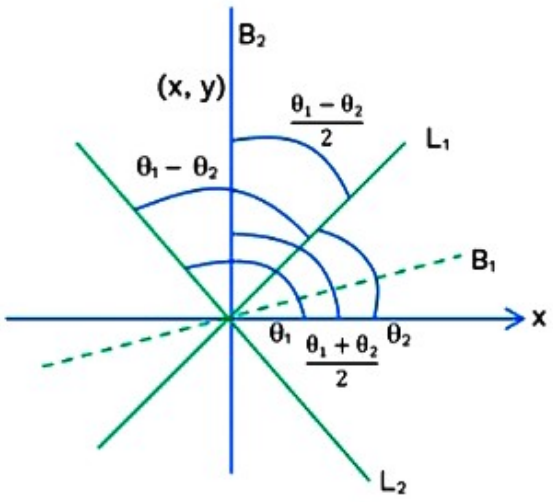
Method-II

$$\frac{\theta_1 + \theta_2}{2} = \theta \Rightarrow \theta_1 + \theta_2 = 2\theta$$

$\tan\theta = y/x$

$\tan2\theta = \tan(\theta_1 + \theta_2)$

$$\frac{2\tan\theta}{1 - \tan^2\theta} = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1\tan\theta_2}$$



$$\frac{2\frac{y}{x}}{1-\frac{y^2}{x^2}} = \frac{m_1+m_2}{1-m_1m_2} = \frac{-\frac{2h}{b}}{1-\frac{a}{b}} = \frac{-2h}{b-a}$$

$$\frac{2xy}{x^2-y^2} = \frac{2h}{a-b} \Rightarrow \frac{x^2-y^2}{a-b} = \frac{xy}{h}$$

Q. **The product of the perpendicular, dropped from (x_1, y_1) to the pair of lines represented by the equation $ax^2 + 2hxy + by^2 = 0$ is?**

Sol. $y - m_1x = 0$

$y - m_2x = 0$

$$\frac{y_1-m_1x_1}{\sqrt{1+m_1^2}} \cdot \frac{y_1-m_2x_1}{\sqrt{1+m_2^2}} = \frac{y_1^2 - m_2x_1y_1 - m_1x_1y_1 + m_1m_2x_1^2}{\sqrt{1+m_2^2+m_1^2+m_1^2m_2^2}}$$

$$\frac{y_1^2 + x_1y_1\frac{2h}{b} + \frac{a}{b}x_1^2}{\sqrt{1+\left(\frac{-2h}{b}\right)^2 - \frac{2a}{b} + \frac{a^2}{b^2}}} = \frac{y_1^2 - x_1y_1(m_1+m_2) + m_1m_2x_1^2}{\sqrt{1+(m_1+m_2)^2 - 2m_1m_2 + m_1^2m_2^2}}$$

$$= \frac{ax_1^2 + 2hx_1y_1 + by_1^2}{\sqrt{(a-b)^2+4h^2}}$$

Note:

Any second degree curve through the four points of intersection $f(x, y) = 0$ and $xy = 0$ is given by $f(x, y) + \lambda xy = 0$ where $f(x, y)$ is also a second degree curve.

Q. **Find equation of angle bisector of straight lines $xy = 0$**

Sol. $\because a - b = 0$

$\therefore$ equation of angle bisector is $x^2 - y^2 = 0$

$\Rightarrow x = y, x = -y$

Q. **Find equation of angle bisector of straight lines $x^2 - y^2 = 0$**

Sol. $\because$ equation of angle bisector $xy = 0$

$\Rightarrow x = 0, y = 0$

Homogenisation:

Combined equation of pair of lines joining origin and the points intersection of a curve and a line

Curve: $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ .... (i)

Line: $lx + my + n = 0$ .... (ii)

$lx + my = -n$ .... (iii)

Let A and B be the points of intersection of the curve and the line

In order to make homogeneous,

$ax^2 + by^2 + 2hxy + (2gx + 2fy)\left(\frac{\ell x + my}{-n}\right) + c\left(\frac{\ell x + my}{-n}\right)^2 = 0$ .... (iv)

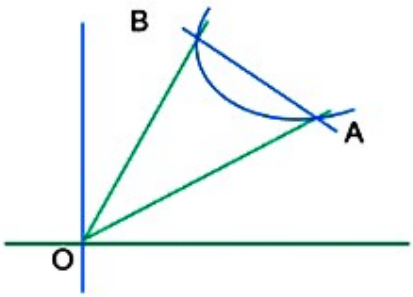
Coordinate of point of intersection of (i) and (ii) will satisfy (iv)

Since it is the combination of (i) and (ii)

Thus, (iv) is a locus of point passes through A and B.

Since (iv) is a homogeneous equation in x and y hence it represents two straight line passing through the origin.

Hence (iv) is a joint equation of OA and OB.



Q. **Find the equation of the line pair joining origin and the point of intersection of the line $2x - y = 3$ and the curve $x^2 - y^2 - xy + 3x - 6y + 18 = 0$. Also find the angle between these two lines.**

Sol. Equation using homogenization

$$x^2 - y^2 - xy + 3x\left(\frac{2x - y}{3}\right) - 6y\left(\frac{2x - y}{3}\right) + 18\left(\frac{2x - y}{3}\right)^2 = 0$$

$\Rightarrow x^2 - y^2 - xy + 2x^2 - xy - 4xy + 2y^2 + 2(4x^2 - 4xy + y^2) = 0$

$\Rightarrow 11x^2 - 14xy + 3y^2 = 0$

angle between lines

$$\theta = \tan^{-1}\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right| = \tan^{-1}\left|\frac{2\sqrt{49 - 33}}{14}\right| = \tan^{-1} 4/7$$

Q. **Find the value of 'm' if the lines joining the origin to the points common the $x^2 + y^2 + x - 2y - m = 0$ & $x + y = 1$ are at right angles.**

Sol. Equation of required pair of SL and

$x^2 + y^2 + x(x + y) - 2y(x + y) - m(x + y)^2 = 0$

$\Rightarrow (2 - m)x^2 - (1 + 2m)xy - (1 + m)y^2 = 0$

$\because$ lines are $\perp$ hence

$a + b = 0$

$\Rightarrow (2 - m) - (1 + m) = 0 \Rightarrow m = ½$

Q. **Show that all chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ subtending right angles at the origin pass through a fixed point find also co-ordinates of the fixed point.**

Sol. Let equation of chord $ax + by = 1$ ... (1)

then required pair of Sls is

$3x^2 - y^2 - 2x(ax + by) + 4y(ax + by) = 0$

$(3 - 2a)x^2 + (4a - 2b)xy + (4b - 1)y^2 = 0$

$\because$ lines are $\perp$ hence

$(3 - 2a) + (4b - 1) = 0$

$2a - 4b = 2 \Rightarrow a - 2b = 1$ ... (2)

by comparison of (1) and (2)

$\Rightarrow$ fixed point $(1, -2)$

Q. **A Line L passing through the point (2,1) intersects the curve $4x^2+y^2-x+4y-2=0$ at the points A, B. If the lines joining origin and the points A, B are such that the co-ordinate axes are the bisectors between them then find the equation of line L.**

Sol. Line passing through (2, 1) is

$(y-1) = m(x - 2) \Rightarrow mx - y = 2m - 1$

now equation of required pair of SL

$$4x^2 + y^2 - x\left(\frac{mx - y}{2m - 1}\right) + 4y\left(\frac{mx - y}{2m - 1}\right) - 2\left(\frac{mx - y}{2m - 1}\right)^2 = 0$$

$\because$ co-ordinate axes are angle bisectors

hence $h = 0 \cong$ or coeff. of $xy = 0$

$$\frac{1}{2m - 1} + \frac{4m}{2m - 1} + \frac{4m}{(2m - 1)^2} = 0$$

$(4m + 1)(2m - 1) + 4m = 0$

$8m^2 + 2m - 1 = 0$

$(4m - 1)(2m + 1) = 0 \Rightarrow m = \frac{1}{4}, -\frac{1}{2}$

equation of line $\frac{x}{4} - y = \frac{1}{2} - 1 \Rightarrow x - 4y + 2 = 0$

$-\frac{x}{2} - y = -1 - 1 \Rightarrow x + 2y = 4$

Q. **A straight line is drawn from the point (1,0) to intersect the curve $x^2 + y^2 + 6x - 10y + 1 = 0$ such that the intercept made by it on the curve subtend a right angle at the origin. Find the equation of the line L.**

Sol. Line passing through (1, 0),

$y = m(x - 1) \Rightarrow mx - y = m$

Equation of pair of SL by homogenization

$$x^2 + y^2 + 6x\left(\frac{mx - y}{m}\right) - 10y\left(\frac{mx - y}{m}\right) + \left(\frac{mx - y}{m}\right)^2 = 0$$

$\because$ lines are $\perp \Rightarrow a + b = 0$

$$8 + \left(1 + \frac{10}{m} + \frac{1}{m^2}\right) = 0, \left(9 + \frac{10}{m} + \frac{1}{m^2}\right) = 0$$

$\Rightarrow m = -1, -1/9$, equation of line $- x - y = -1 \Rightarrow x + y = 1$

$\frac{-x}{9} - y = \frac{-1}{9} \Rightarrow x + 9y = 1$